

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics

In partial fulfillment for the requirements in Capstone 1 and
Research Subject

Fernandez, Criss Angiela - BSIT 3
Mandado, Keziah - BSIT 3
Marcelo, Ziny Jen - BSIT 3
Suamen, Maximo - BSIT 3
Suficiencia, Joshua - BIST 3

Data Informatics Track

Second
Semester
2025-26

TABLE OF CONTENTS

CHAPTER 1.....	5
1.1 Background of the Study.....	5
1.2 Statement of the Problem.....	14
General Problem Statement.....	14
Specific Statement of the Problem.....	15
1. Customer-Intake and Ordering Inefficiencies.....	15
2. Operational, Scheduling, and Production Disruptions.....	19
3. Data privacy and bookkeeping risks.....	21
1.3 Objectives of the Study.....	23
General Objective.....	23
Specific Objectives.....	24
1.4 Scope and Delimitation.....	26
1. System Environment and Accessibility.....	26
2. User Roles and Access Levels.....	27
3. Scope of the Customer Portal.....	28
4. Scope of the Administrative and Kitchen Dashboard.....	30
5. Delimitation of the Study.....	32
1.5 Significance of the Study.....	34
1. Bakery Customers and Guests (The Buyers).....	35
2. Administrative Control Personnel (Bakery Staff and Decorators).....	36
3. Executive Management (The Bakery Owner).....	37
4. The Organization (C&N Bakery as a Business).....	39
5. Future Systems Developers.....	40
6. Future Academic Researchers.....	40
7. The Local Food Service Industry.....	41
8. BSIT Students.....	41
1.6 Definition of Terms.....	42
1. Web-Based System.....	42
2. User Authentication and Account Access Control.....	43
3. Interactive Custom Cake Builder.....	44
4. Automated Pricing Calculator.....	44
5. Online Menu Catalog.....	45
6. Private Customer Dashboard and Progress Tracker.....	46
7. Sales Analytics Charts.....	47
8. Menu Availability Toggle Control.....	47
9. Auto lock booking calendar.....	48
CHAPTER 2.....	49
2.1 Foreign Studies.....	49
2.2 Local Studies.....	63

2.3 RELATED LITERATURE.....	76
2.4 Synthesis of the Study.....	92
CHAPTER 3.....	96
3.1 Research Design.....	96
1. Descriptive Research.....	96
2. Developmental Research.....	99
3. Why We Did Not Use Experimental Research.....	101
3.2 System Development Methodology.....	104
1. Requirements Gathering.....	106
2. Design.....	108
3. Implementation.....	110
4. Testing.....	112
5. Deployment.....	114
6. Maintenance.....	115
3.3 System Architecture.....	117
1. The Client-Side Application.....	118
2. The Backend-as-a-Service (BaaS) Layer.....	118
A. Firebase Authentication.....	119
B. Firebase Realtime Database.....	119
3.4 Tools and Technologies Used.....	121
1. HyperText Markup Language (HTML5).....	121
2. Cascading Style Sheets (CSS3).....	122
3. Vanilla JavaScript.....	123
4. Firebase Authentication.....	124
5. Firebase Realtime Database.....	124
6. Firebase Backend Infrastructure (No-Cost Spark Plan).....	125
7. Visual Studio Code (VS Code).....	126
8. Git and GitHub.....	127
9. Firebase Console Dashboard.....	128
3.5 Data Gathering Techniques.....	129
1. Interviews.....	129
2. Observation.....	130
3. Document Analysis.....	132
3.6 Testing Procedures.....	135
1. Unit Testing.....	135
2. Integration Testing.....	137
3. System Testing.....	138
3.7 Evaluation Criteria.....	141
1. Functionality.....	141
2. Usability.....	142
3. Reliability.....	143

4. Efficiency.....	144
5. Security.....	145
CHAPTER 4.....	147
4.2 Overview of the Developed System.....	147
Figure 4.1 illustrates the overall operational workflow of the developed system.....	149
4.3 System Development Methodology.....	153
Requirements Gathering.....	154
System Planning.....	155
System Design.....	156
Software Development.....	156
Testing.....	157
Deployment.....	157
Maintenance.....	158

CHAPTER 1

1.1 Background of the Study

In today's fast-paced world, businesses across all sectors are shifting to online platforms to keep up with the growing expectations of their consumers. The food service and enterprise baking industry is no exception to this trend. As tech-savvy individuals increasingly turn to digital applications for seamless, on-demand transactions, local micro-enterprises that continue to rely on manual, analog processes face significant challenges. Traditional, non-automated methods often result in critical operational inefficiencies, reduced market reach, and an inability to meet contemporary consumer demands. This shift in buyer behavior has turned digital convenience from a luxury into an absolute necessity for survival in the food retail industry. When small stores lack a modern web presence, they silently lose customers to larger competitors who offer instant online ordering.

The underlying problem stems from how consumer data is collected and managed during initial touchpoints. In a typical digital commerce environment, data inputs are structured, validated, and processed instantaneously by backend application logic.

Conversely, traditional store structures rely heavily on human

memory and fragmented paperwork to handle transactions. This lack of automated validation means that business transactions are highly susceptible to data loss, tracking delays, and operational blind spots. For community-scale enterprises, continuing to operate within a purely analog framework creates a technical barrier that cuts them off from the modern digital marketplace, making it increasingly difficult to sustain consistent growth or build long-term customer loyalty.

C&N Bakery is a local bakery operating within a highly competitive local market, offering a wide selection of baked goods categorized into standard product groups, including various daily pastries and traditional baked items. In addition to mass retail options, the establishment specializes in bespoke, high-value custom cake orders and supports a pre-ordering framework designed for clients requiring specialized baked goods for celebrations, weddings, debuts, and milestones. Despite managing a diverse product line and catering to a growing clientele, C&N Bakery handles its entire business workflow through manual mechanisms, such as physical walk-ins, phone calls, and scattered social media messages, without any centralized digital infrastructure to sustain its operations. This lack of a unified system means information is scattered across different mediums, making daily tasks stressful,

time-consuming, and disorganized.

When a business does not use a centralized repository to store its records, its information easily gets scattered across too many different places. At C&N Bakery, a single custom transaction often leaves a fragmented data trail across multiple communication channels, where design layouts, delivery deadlines, and payment details are recorded on completely unlinked mediums. Because these data points are isolated, the staff must manually piece together the customer's requirements before coordinating production. This approach consumes massive amounts of time and makes day-to-day operations chaotic, turning simple routine tasks into exhausting manual procedures that drain the small business's daily energy and workforce productivity.

Managing custom cake orders is inherently more complex than selling regular pastries over the counter. A custom cake order requires capturing and tracking multiple precise design dimensions, flavor variations, tier structures, color schemes, and stylistic preferences simultaneously. On top of that, the bakery must carefully coordinate the exact pickup or delivery fulfillment dates to guarantee product freshness and structural stability during transport. Because C&N Bakery relies on handwritten notebooks, loose paper receipts, and unorganized

chat histories, it is incredibly easy for staff to forget or misinterpret these specific design details by the time production begins. Plus, when workers try to schedule these orders by writing them down in a physical logbook, human error often causes accidental double-bookings, overloading the kitchen with more volume than the baking team can execute within a single working shift.

This operational vulnerability is deeply tied to the technical concept of data mutation. When a customer outlines their custom selections through casual communication channels, that information exists as an unformatted text string. As that request moves from the front counter staff to the production area, the information is manually rewritten onto paper order tags. Every time a human manually copies, interprets, or rewrites a data string, the higher the risk of transcription mistakes occurring. Essential design attributes can easily be misread, miscalculated, or completely forgotten during the transfer. This manual workflow creates a weak system loop where the original intent of the customer rarely matches the actual physical instructions given to the baking staff on the kitchen floor.

When the bakery relies on scattered chats and unorganized paper notes, it creates a high risk of transcription mistakes during

the order-taking process. These data entry errors directly lead to incorrect specifications being sent to the kitchen. Once a product is baked with the wrong design or configuration, it triggers a chain reaction of operational failures across the business. This workflow breakdown results in wasted raw ingredients, disrupted baking queues, and unrecoverable financial losses for the bakery, while severely damaging long-term customer trust and brand reputation.

The financial damage caused by these data errors is made worse by the rigid nature of the baking process itself. Unlike other food service businesses that can quickly prepare a replacement dish within minutes, a custom cake requires hours of uninterrupted preparation, baking, cooling, structural stabilization, and detailed artistic application. If a configuration error is discovered late in the production line due to a manual data entry mistake, the entire product must often be discarded. This means the bakery loses both the expensive ingredients used and the valuable time slots in the production schedule. This disruption delays every other order waiting in the queue, creating a stressful work environment and causing a backlog of delayed fulfillment timelines that ripples through the entire business day.

The operational difficulties are further worsened by the rapidly

evolving commercial ecosystem in the Philippines, marked by the widespread adoption of digital transactions, online banking, and electronic wallet applications. Modern consumers expect seamless transaction flows that offer immediate pricing feedback, cost transparency, and structured access to basic business policies, combined with the convenience of settling accounts remotely. For C&N Bakery, maintaining a purely manual setup limits its ability to engage this expanding, digitally native market segment that prioritizes efficiency, speed, and remote ordering. When a customer has to wait hours for a manual message reply just to get a basic price quote, they often abandon the purchase entirely.

This bottleneck is a direct result of operating without an automated computational engine to handle basic pricing calculations. In a manual communication model, every customer inquiry regarding custom options forces a staff member to stop their physical kitchen tasks to manually calculate costs line-by-line. This slow process creates a massive communication delay, leaving potential buyers waiting in digital queues for simple information. In the modern local economy, where consumers are accustomed to instant digital responses, these long waiting times cause high cart abandonment rates, as buyers quickly abandon the conversion funnel to buy from larger automated

competitors instead.

Beyond customer-facing constraints, the absence of a unified web platform introduces severe security risks and data management obstacles. On the user side, clients lack a secure, authenticated interface to manage their personal records or securely monitor the status of their active orders. For administrative personnel, operating without a secure, centralized administrative console means that confidential business information—such as historical sales records, client contact logs, daily revenue metrics, and sensitive operational statistics—remains vulnerable to data corruption, unbacked physical destruction, and unauthorized exposure, making data integrity a constant corporate concern. If a single paper logbook is misplaced, damaged, or stained in the kitchen, the bakery completely loses its financial records and customer transaction histories.

This total reliance on single-copy physical storage devices exposes the business to immense operational hazards. Paper logbooks kept inside an active kitchen are constantly exposed to water damage, grease stains, heat degradation, and accidental loss. From an information security perspective, storing a company's entire transaction history on unsecured paper slips leaves confidential client addresses and contact numbers open to

unauthorized eyes. Furthermore, physical records offer zero data recovery options; if an environmental hazard occurs in the shop, years of vital business history are permanently destroyed in minutes, leaving management with no backup logs to reconstruct their operational data.

These combined systemic vulnerabilities generate persistent issues for both key stakeholder groups. Customers endure a fragmented user experience due to the lack of a centralized online catalog with dynamic filters to browse products, or an intuitive mechanism to submit custom reference visual designs and decorative specifications. Concurrently, the production and administrative staff struggle to monitor simultaneous orders across various stages of readiness, which leads to scheduling errors, missed deadlines, and fulfillment delays. Without a visual tool to track order states, progress, or daily capacities, the small kitchen team easily becomes overwhelmed during peak holiday seasons and busy weekend ordering queues.

This communication gap creates a significant operational barrier inside the production facility. Because the front counter staff and the production team do not share a synchronized information system, they easily lose track of order priorities and preparation sequences. The production staff has no seamless way of identifying urgent queues, and the front counter staff cannot

give customers accurate updates on when their orders will be ready. This forces everyone to constantly stop their physical duties to verbally verify order statuses. This constant workplace interruption wastes valuable labor time, stresses out the staff, and leads to scheduling mistakes that leave customers waiting and frustrated.

To address these operational vulnerabilities and support the digital transition of the business storefront, this study proposes the development of the **Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics for C&N Bakery**. The proposed web-based application aims to establish a centralized digital infrastructure engineered to streamline order acquisition, dynamically balance daily kitchen capacity limits, track production stages across a clear lifecycle, and compile historical business records into structured sales reports. By replacing vulnerable, manual mechanisms with a secure, database-driven workspace, the project seeks to eliminate data transcription errors, safeguard operational data, optimize kitchen productivity, and provide an efficient, transparent web presence tailored specifically to the operational needs of C&N Bakery.

1.2 Statement of the Problem

General Problem Statement

The primary challenge facing C&N Bakery is its total reliance on manual operational workflows, physical paper logbooks, and fragmented social media chat threads to manage daily business activities. Because the establishment lacks a centralized, database-driven web platform to synchronize data across departments in real time, operational records remain severely fragmented. Essential transaction variables—including intricate custom cake specifications, fulfillment schedules, dynamic menu updates, raw material availability, and confidential customer records—are distributed across loose paper slips, handwritten kitchen calendars, and unindexed social media messaging histories.

Relying on this unlinked, non-automated setup makes the business highly vulnerable to data transcription errors, miscommunication, and permanent file destruction. Without an automated digital framework to systematically capture, validate, and lock in order specifications upon checkout, the bakery operates without a single, verified source of truth for its daily transactions. On a daily level, this broken operational workflow leads to severe communication backlogs during design consultations, frequent product assembly errors on the kitchen

floor, costly ingredient waste, and accidental kitchen overbookings. Furthermore, storing financial records, payment logs, and sensitive customer contact information in unsecured physical notebooks exposes the bakery to severe privacy vulnerabilities, regulatory non-compliance, and data loss.

Specific Statement of the Problem

To address the general problem, the specific challenges of C&N Bakery are categorized into three major problems:

1. Customer-Intake and Ordering Inefficiencies

- **Communication Backlogs and Delayed Responses:**

Customers who want to browse the current menu, check available flavor variations, verify daily product stocks, or ask about custom pricing are forced to call the shop directly or send private messages through social media platforms. Because customer communication relies entirely on manual, one-by-one text replies from front-desk workers and baking staff who are simultaneously busy mixing batter, decorating cakes, or managing walk-in counter queues, the bakery experiences severe message backlogs during high-volume periods, holidays, and weekends. It is common

for potential buyers to wait several hours—or even until the next business day—just to receive a basic price quote or check if a specific calendar date is open. This response delay creates immediate transactional barriers, frustrating interested buyers during critical decision-making moments and causing high customer drop-off rates. Ultimately, when **potential** clients face these long waiting times, they routinely abandon the inquiry process entirely and take their business to faster, automated online competitors.

- **Pricing Guesswork and Math Errors:**

The staff currently determines prices right in the middle of active messaging conversations with customers by relying on quick mental math, rough estimates, or scanning through loose, unlinked physical price sheets. This informal approach frequently leads to inconsistent pricing quotes for nearly identical cake designs, human calculation mistakes, and a total lack of itemized cost transparency for the buyer. Because there is no structured pricing calculator available to the customer, the buyer cannot see how modifying specific design parameters—such as adding an extra cake tier, choosing a premium flavor filling, requesting intricate fondant decorations, or setting an urgent fulfillment deadline—directly affects their total

bill. This lack of clear, real-time cost breakdowns creates consumer hesitation, leads to endless back-and-forth price negotiations, and ultimately damages customer trust during the ordering process.

- **Misunderstood Cake Designs and Data Transcription Errors:**

Staff members must regularly read through long, unstructured chat histories across social media apps to find a customer's custom design requests and manually copy those details onto physical paper kitchen slips. This manual hand-off creates a high risk of missing, misreading, or miscopying critical cake specifications—such as exact color themes, tier dimensions, frosting types, structural support needs, specialty fondant toppers, and custom written inscriptions. Because there is no structured form or digital checklist to capture these details during the intake phase, important preferences frequently get buried in casual messaging conversations. When these design details are misread or forgotten during the transfer to the production floor, it leads to incorrect cake assemblies in the kitchen. Ultimately, these transcription mistakes result in client rejections upon pickup or delivery, urgent last-minute re-work, and a heavy waste of expensive baking ingredients that directly eats into the business's profits.

- **Information Gaps and Customer Hesitation:**

C&N Bakery's manual setup offers no public, self-service for basic store policies, disclaimers, lead-time rules, or payment guides. Customers seeking basic operational details—such as required order advance times, refund or cancellation terms, nut and dairy allergen warnings, or step-by-step digital payment instructions—are forced to message the staff directly to ask about them. This lack of clear, readily accessible information creates unnecessary communication backlogs, forcing workers to answer the same repetitive questions over and over again. More importantly, it causes immediate transaction hesitation during the final stages of ordering, as prospective buyers are often reluctant to send down-payments or finalize high-value custom orders without seeing clear, official shop guidelines provided upfront.

2. Operational, Scheduling, and Production Disruptions

- **Kitchen Overbooking and Overlapping Deadlines:**

Fulfillment deadlines are currently tracked using a single physical wall calendar hung in the kitchen area. Because multiple employees accept customer orders simultaneously across direct phone calls, walk-in counter visits, and

unlinked social media messaging channels, workers lack a shared, real-time view of the daily production schedule. As a result, staff members routinely book orders without knowing that the kitchen has already reached its maximum workload capacity for a specific date or time slot. This lack of centralized schedule management regularly leads to accidental overbookings and overlapping fulfillment deadlines. On busy days, this overbooking severely overloads the kitchen team, forcing bakers to rush through complex preparation steps, compromising delicate cake decorating quality, and causing delayed customer pickups that harm the bakery's reputation.

- **Disorganized Production Tracking and Status Blindness:**

The kitchen staff does not have a centralized, real-time method to monitor where open orders stand within the preparation lifecycle. Instead of using a shared visual board or tracking system, employees are forced to yell across the noisy kitchen room to ask about progress or constantly pause work to flip through stacks of loose paper receipts. This lack of visual order tracking creates significant operational confusion , as bakers and decorators must repeatedly stop their detailed, hands-on tasks just to answer status inquiries from front-counter

workers who are trying to update waiting walk-in clients or online messaging inquiries. These constant workplace interruptions disrupt the baking team's focus, slow down the overall production pipeline, and make it difficult to identify which urgent orders need to be prioritized during high-volume work shifts.

- **Logistical Misalignments and Delivery Mix-ups:**

Because pickup schedules, delivery addresses, and customer contact criteria are written down on loose, unlinked paper documentation, it is incredibly easy for kitchen staff to misread target fulfillment dates or mix up address notes. Without a systematic coordination method to manage dispatch schedules, a baker might accidentally prepare and decorate a delicate custom cake days before the customer actually needs it. This causes the fresh product to sit in storage, losing its freshness, and visual appeal before pickup. Conversely, staff members might overlook a delivery slip entirely, realizing an order is due only when the customer arrives or sends a message, leading to severe fulfillment delays, chaotic emergency decor work, wrong delivery addresses, and uncoordinated logistics failures that damage customer trust.

- **Slow Hand-offs and Manual Record Keeping:**

Confirming that an order was successfully prepared, paid for, and handed over to the buyer requires manual verbal follow-ups and physical file logging at the counter. If a tired worker forgets to file a completed order slip into the physical binder ledger book at the end of a long, exhausting shift, that entire sales record effectively vanishes from the day's bookkeeping. This missing data creates immediate financial blind spots and makes it impossible to verify if the order was properly completed or if cash balances match physical inventory.

3. Data privacy and bookkeeping risks

- **Unsecured Business Records and Lack of User Privacy:**

Customers lack a private interface where they can safely access their accounts, while confidential business financial logs and customer databases are kept on unbacked paper notebooks or personal mobile phones, leaving sensitive business data exposed to physical theft, damage, or unauthorized exposure.

- **Slow Payment Tracking and Accounting Errors:**

Managing down-payments and cash balances using manual notebooks is highly labor-intensive and prone to math mistakes. Because proof-of-payment screenshots from digital wallets are kept loose in mobile phone photo galleries, matching a payment slip to a specific paper order requires tedious manual cross-checking, causing frequent money discrepancies.

- **Absence of Business Insights and Data Blindness:**

The bakery owner cannot easily check if the business is growing or calculate profit margins without spending hours doing manual math at the end of every month. To see how the bakery is doing, the owner must manually sort through and add up hundreds of individual physical order slips line-by-line, leaving management unable to track changing customer preferences or predict busy seasonal rushes.

1.3 Objectives of the Study

General Objective

The primary objective of this study is to design, develop, and implement a secure, web-based ordering, smart scheduling, and sales analytics platform for C&N Bakery to transition its manual transactional workflows into a shared online platform. This study aims to replace paper-dependent habits—such as manual logbooks, physical kitchen calendars, and fragmented communication threads—with an integrated platform that shares data access across customers, kitchen personnel, and management. Ultimately, the system seeks to eliminate pricing ambiguities through automated cost computation, protect the kitchen from operational overbooking via automated scheduling guards, and provide business analytics to support data-driven decision-making.

Specific Objectives

To successfully achieve the general objective of the study, the research will fulfill the following structured engineering and research milestones:

1. To analyze and document the existing manual workflows of C&N Bakery regarding product ordering, custom cake configuration, kitchen scheduling, and sales record-keeping

to establish the foundational technical requirements of the platform.

2. To plan out how the website will work, map out the database where information is stored, and design clean, easy-to-use screens. This ensures that the shop's data stays accurate, sensitive business details remain password-protected, and the website is simple for everyone to navigate.
3. To build the public website that customers see, combining an online menu that lets users filter pastries by category with a custom cake designer page. This designer setup will automatically calculate and update the estimated cake price on the screen the moment a customer changes their design choices.
4. To build a secure checkout step on the website that lets buyers send their custom text messages and upload design reference photos along with their order. This guarantees the customer's design requests travel safely straight to the shop's database without getting lost or altered.
5. To build an automated scheduling tool that keeps track of how many custom cake orders are placed for a specific date. Once the kitchen hits its daily limit, the system will automatically lock that calendar date on the website so no more customer bookings can be made for that day.
6. To build an order tracking layout for the kitchen staff

that displays all of a customer's cake choices, text notes, and uploaded photos together on a single view. This allows the baking team to see all specifications at once and systematically update an order's progress as it moves from baking to final decorating.

7. To build a reporting section for the owner that automatically gathers past sales records and displays them in clear visual charts. This makes it easy for management to monitor monthly revenue trends and instantly see which menu items are selling the best.
8. To set up secure user login rules and data validation steps across the website. This ensures that customers and staff only see the specific information they are allowed to access, while protecting private business financial logs and preventing typing errors during checkout.
9. To test and evaluate the completed website through structured trial runs with developers and actual users. This process measures how fast the system runs, verifies that the security rules work correctly, and uses standard evaluation feedback to ensure the website is easy and practical for everyone to use.

1.4 Scope and Delimitation

This study focuses on designing, building, and testing a secure web-based ordering, smart scheduling, and sales analytics platform specifically for the operational workflows of C&N Bakery. The primary goal of the system is to transition the bakery's manual, paper-dependent practices—such as handwritten order notebooks, physical kitchen wall calendars, and scattered social media chat conversations—into a single, organized digital platform. The software structure is split into two primary components: a public customer portal for browsing and order submission, and a secure administrative dashboard to optimize kitchen operations, capacity management, and business record-keeping.

1. System Environment and Accessibility

The platform is developed exclusively as a responsive web application, eliminating the need for end-users or staff members to download, install, or update native mobile software. The user interface utilizes a flexible grid layout and media queries that automatically adapt screen components across a wide range of display dimensions, including desktop computers, laptops, tablets, and smartphones. The application is designed to be cross-browser compatible, operating reliably across major modern web browsers—such as Google Chrome, Firefox, Apple Safari. Access to the platform requires a device with an active internet

connection to communicate with the cloud-hosted database server in real time.

2. User Roles and Access Levels

To keep data accurate and secure, the system splits user access into three distinct roles:

- **Bakery Customers:**

Public visitors can freely navigate the platform to view the digital product menu, check item descriptions, and review store policies and disclaimers. Registered and authenticated buyers gain access to a personal account dashboard where they can configure custom cake requests, submit order details and track the progress of their active orders.

Administrative and Staff Dashboard (Secure Backend Portal): This single secure backend is shared by both the bakery owner and kitchen production staff upon logging in.

- **Production & Kitchen View:** Kitchen staff and decorators access this dashboard to review complete order specifications, confirm schedules, inspect customer-uploaded reference photos, update order progress through baking stages, and confirm order pickups or deliveries.

- **Administrative Controls:** The owner uses this same backend dashboard to manage store-wide settings, adjust base menu pricing, toggle product visibility, set daily order capacity limits, and review automated business analytics reports. If role-based restriction features are disabled or kept shared, staff members access the same overall dashboard workspace alongside the owner to facilitate seamless daily multitasking.

3. Scope of the Customer Portal

The customer website includes all the features needed to guide a buyer from browsing the menu to submitting an order:

- **Categorized product catalog:**

The shop organizes items into clear product sections with filtering options, helping users find pastries and local treats quickly.

- **Live price calculator:**

This page guides customers step-by-step through custom requests. As users pick their options—like tiers, sizes, flavors, and frosting—a background script automatically

calculates and updates the estimated baseline price on the screen so pricing is completely transparent.

- **Design reference uploader:**

The order form lets customers attach inspiration photos and type out cake dedication text directly into the system, preventing design and spelling mistakes.

- **Order limit checker:**

During checkout, the customer chooses a fulfillment date on an interactive calendar. Dates that have hit the bakery's maximum daily limit are automatically greyed out by a background scheduling rule. The form then collects necessary pickup or delivery details.

- **Self-Service Order Progress Tracker:**

Logged-in users can check a status tracker in their profile to see exactly what stage their cake is in as the kitchen updates it, eliminating the need to call or text the shop for updates.

4. Scope of the Administrative and Kitchen Dashboard

The secure backend dashboard gives management and the kitchen team full control over daily workflows, schedules, and records:

- **Bakery performance summary:**

Upon logging in, the owner sees an overview of key performance indicators, like total revenue for the month, pending tasks, and active kitchen workloads.

- **Main order directory:**

This section groups all incoming orders into a clean, searchable directory. Staff can filter records by customer details, product types, deadlines, and payment status without touching paper.

- **All in one design summary:**

To avoid mistakes in the kitchen, clicking an active order opens a single layout displaying the customer's uploaded reference image alongside their exact flavor choices and text notes.

- **Step-by-Step Production Status Controller:**

The kitchen view lets bakers smoothly update an order's status as it moves through different preparation stages. Staff can also check and approve payment status after

reviewing customer receipts.

- **Instant menu toggles:**

Staff can turn off the visibility of specific menu items on the backend, which removes out-of-stock pastries from the public storefront instantly during ingredient shortages.

- **Auto booking limiter:**

This tool runs continuously in the background to count incoming orders for each date. The moment a daily limit is met, the system automatically blocks that date from accepting more online reservations.

- **Sales Analytics Visualizer:**

This administrative tool gathers historical transaction records from the database and turns raw numbers into clear charts, highlighting monthly revenue trends and the bakery's most popular products.

5. Delimitation of the Study

To keep the project realistic and ensure it is finished on time, the following features are explicitly excluded from this study:

- **No Native Mobile Applications:**

The project is strictly confined to a responsive web browser design; it does not include developing or publishing standalone mobile apps for Google Play or the Apple App Store.

- **No Direct Financial API Integrations:**

The website does not connect directly to banking systems or automated payment gateways. Payment verification is handled manually by the staff, who review uploaded receipt screenshots and manually mark the order as paid.

- **No Automated Third-Party Logistics Routing:**

The application works completely independently of external courier networks. It does not link with delivery services (like Grab or Lalamove); setting up drivers and managing delivery dispatch must be done by the staff outside the system.

- **No Raw Material Inventory Tracking:**

The system does not track raw baking ingredients (like the weight of remaining flour, sugar, or eggs). Inventory management relies entirely on staff observations and using the manual storefront visibility toggles.

- **No Human Resource or Payroll Systems:**

The platform is built solely for orders and scheduling. It does not include employee timecards, shift scheduling, or payroll management features.

- **No Supply Chain or Vendor Logs:**

The software lacks tools for managing raw ingredient suppliers, creating wholesale purchase orders, or tracking fluctuating ingredient costs from vendors.

- **No Offline Operational Capability:**

The application depends completely on a live database server. It does not support offline mode or offline data saving; users must have a working internet connection to use the website.

1.5 Significance of the Study

Building the Custom Cake Ordering Platform with Smart Scheduling

and Sales Analytics brings great practical value to everyone involved in the daily operations and growth of C&N Bakery. As the local food and bakery industry shifts toward digital spaces, traditional neighborhood bakeshops must update their tools to stay relevant and keep up with modern competitors. This study serves as a real-world demonstration of how a local bakery can successfully transition away from the stress and physical disorganization of manual paperwork. By replacing easily lost handwritten logbooks, fading sticky notes, and scattered social media message threads with a single, all-in-one web application, this project creates a smooth, dependable ordering process. This new system ensures absolute pricing honesty for everyday customers, protects the kitchen team from accidental booking overlaps, and safeguards historical business records from getting physically damaged or lost. By modernizing these daily tasks, this study provides clear, long-term benefits to a wide group of stakeholders:

1. Bakery Customers and Guests (The Buyers)

Traditionally, everyday buyers trying to order custom cakes or local delicacies from neighborhood shops face frustrating customer service ordering roadblocks, including scattered chat histories, delayed responses, and unclear pricing structures. The launch of this online platform directly assists buyers by

providing:

- **Honest and Instant Pricing:**

Customers benefit from complete price transparency, receiving real-time cost feedback as they select different cake configurations and design complexities, which removes pricing guesswork and transactional anxiety.

- **Accurate Design Submission:**

The platform allows buyers to seamlessly submit their design reference photos alongside their flavor preferences and structural specifications, minimizing design misunderstandings and ensuring their expectations match the final product.

- **Independent Order Tracking:**

Users gain access to real-time status updates regarding their active orders directly from the bakery, eliminating the need to wait for manual staff replies to confirm the current production phase.

- **Convenient Menu Browsing:**

The centralized digital catalog allows customers to comfortably browse offerings and plan their celebrations at

any time from any internet-connected device.

2. Administrative Control Personnel (Bakery Staff and Decorators)

Kitchen staff and cake decorators working within a manual setup operate under immense daily pressure to track upcoming deadlines and unique design themes using fragile paper records. This project introduces organization to the kitchen workspace by providing:

- **Workload Protection:**

The platform prevents accidental overbooking by regulating the daily volume of custom orders, protecting the kitchen team from exhausting workloads and operational burnout.

- **Production Accuracy:**

Cake decorators can review complete design instructions and reference images within a single, unified view, eliminating errors caused by searching through loose papers or disjointed chat threads.

- **Step-by-step production stages:**

The kitchen interface allows baking personnel to smoothly

advance orders through verified preparation milestones, moving a cake from mixing to baking and final decorating.

- **Efficient Menu Management:**

Staff can update product availability instantly, preventing customers from ordering and paying for treats that are currently out of stock due to raw ingredient shortages.

3. Executive Management (The Bakery Owner)

For the business owner, managing a specialized shop without a centralized digital tool complicates financial auditing and long-term business planning. This platform streamlines daily oversight by providing:

- **Automated Financial Records:**

The system organizes transaction histories, down-payments, and payment verifications into a reliable digital ledger, reducing the risk of human calculation errors.

- **Visual Sales Insights:**

The owner gains access to automated performance tools that track long-term business growth and highlight top-selling

products, facilitating data-driven inventory and financial planning.

- **Data Protection and Privacy:**

Secure role-based access rules ensure that sensitive financial metrics and customer details are restricted solely to the owner, keeping business records safe from unauthorized eyes.

- **Strategic Business Decisions:**

By providing unalterable digital summaries of customer buying trends, the platform helps the owner make smarter choices regarding ingredient budgeting and seasonal menu updates.

4. The Organization (C&N Bakery as a Business)

Transitioning to an automated platform professionalizes the C&N Bakery brand and protects its commercial reputation from the fallout of missed deadlines or forgotten details. The business benefits from:

- **Prevention of Revenue and Resource Waste:**

Enforcing systematic confirmation and tracking parameters

ensures the kitchen never wastes costly ingredients on unverified custom orders.

- **Stronger Local Competitiveness:**

The platform grants a small, family-owned shop a massive competitive edge by offering the modern digital conveniences that tech-savvy buyers expect today.

- **Scalable Business Growth:**

By streamlining internal workflows and automating communications, the bakery can safely handle higher order volumes during peak holiday seasons without overextending its administrative capabilities.

5. Future Systems Developers

For software engineers building e-commerce applications for specialized food markets, standard online stores lack the strict booking limits and custom configuration tools required for custom baking. This research assists developers by offering:

- **Custom order reference:**

This study serves as a practical guide for building custom product forms that calculate prices in real time as users

click different options.

- **Booking Rules Framework:**

This study provides a practical template for setting up automated calendar boundaries that prevent system overbooking based on fixed daily shop capacities.

6. Future Academic Researchers

This study provides valuable documentation for researchers analyzing how small businesses transition from manual operations to localized web platforms. It provides documented local business workflows that can be directly compared to other retail models and serves as a foundational reference for designing systems for made-to-order businesses. Furthermore, it outlines structured paths for future student projects, such as integrating automated SMS notifications or third-party logistics applications.

7. The Local Food Service Industry

The success of this platform demonstrates that practical, high-quality digital solutions are accessible to neighborhood bakeshops and small restaurants without requiring massive corporate budgets. Raising local service standards through

transparent pricing and digital reliability encourages nearby food vendors to modernize their operations, elevating the industry as a whole.

8. BSIT Students

For Information Technology students, this documentation serves as a practical blueprint for breaking down manual business problems into functional technical features. It demonstrates how to translate real-world storefront operations into clean database states and systematic workflows, illustrating the core principles of effective user interface design and role-based data security.

1.6 Definition of Terms

To ensure a clear and accurate understanding of the technical concepts, system operations, and structural components discussed throughout this study, the following key terms are defined both conceptually (how they operate in general) and operationally

(how they are applied inside the C&N Bakery platform):

1. Web-Based System

- **Conceptual Definition:**

An application software architecture that runs on a remote server and is accessed exclusively via a standard web browser over an internet connection, requiring no local installation or hard drive storage on the user's local machine.

- **Operational Definition:**

This refers to the online software platform engineered specifically for C&N Bakery. It allows customers, cake decorators, and the business owner to access the retail menu, place cake pre-orders, monitor tracking statuses, and evaluate monthly financial metrics using any available internet-connected device—such as a smartphone, tablet, laptop, or desktop computer—without having to download an application file from an app store.

2. User Authentication and Account

Access Control

- **Conceptual Definition:**

A digital security gatekeeping protocol that verifies a user's identity through login credentials and checks their pre-assigned database permissions before granting access to protected sections of a network.

- **Operational Definition:**

This functions as the system's security checkpoint during the login sequence. By reading the user's account type, the system automatically routes the user to their appropriate environment. This ensures that everyday customers are restricted to the public ordering pages, kitchen staff are directed only to production workflows, and the bakery owner maintains exclusive, password-locked access to sensitive financial sales records.

3. Interactive Custom Cake Builder

- **Conceptual Definition:**

A dynamic front-end web portal interface that allows

customers to visually select, configure, and modify specific design elements of a product prior to finalizing a transaction.

- **Operational Definition:**

This is the dedicated configuration page on the website where customers design a custom cake from scratch. Users can select a guided sequence of design variables—including the total number of tiers, layer sizes, sponge flavors, and frosting variations—while simultaneously typing written text inscriptions and uploading visual design reference photos.

4. Automated Pricing Calculator

- **Conceptual Definition:**

A client-side computing script that runs in the background of an interface to instantaneously compute arithmetic totals the exact millisecond a user alters an input variable on the screen.

- **Operational Definition:**

This is the mathematical background script operating inside the Interactive Custom Cake Builder page. The instant a

customer changes a design option, such as upgrading a cake tier size or picking a premium frosting flavor, the script recalculates the baseline cost metrics and updates the final price quote on the user's screen in real time, removing the need for manual price quotes via phone messages.

5. Online Menu Catalog

- **Conceptual Definition:**

A structured digital storefront layout that organizes a commercial business inventory into searchable, filterable categories to ease online navigation.

- **Operational Definition:**

This serves as the public digital browsing showcase for C&N Bakery's full baked product line. It automatically separates items into distinct product category views—such as customized cakes, cupcakes, kakanin, and sapin-sapin—and provides a text-based search input alongside quick budget filters so users can find specific treats that fit their spending limits.

6. Private Customer Dashboard and Progress Tracker

- **Conceptual Definition:**

A restricted user profile workspace that displays personalized account data, transaction history lists, and real-time operational status updates regarding active requests.

- **Operational Definition:**

This is the private account hub accessible only to registered and logged-in C&N Bakery customers. Within this workspace, users can safely manage their account details, view past transaction records, and monitor a live visual timeline that reads data directly from the kitchen to show if their active order is currently marked as Preparing, Baking, Packing, or out for Delivery.

7. Sales Analytics Charts

- **Conceptual Definition:**

A data visualization module that parses archived transaction logs from a relational database and charts them

into graphical formats to highlight operational trends.

- **Operational Definition:**

This reporting system automatically gathers completed order histories and transforms the raw numbers into visual data tools: a Monthly Revenue Graph that tracks long-term business growth, and a Top-Selling Products Graph that visually highlights which items, such as traditional kakanin or custom birthday cakes, are generating the most revenue.

8. Menu Availability Toggle Control

- **Conceptual Definition:**

A backend binary administrative switch used by system operators to instantly modify the visibility status of a product row within a public-facing database catalog.

- **Operational Definition:**

This refers to the digital availability controls positioned next to every individual pastry listing in the admin management panel. If the kitchen runs out of the raw ingredients required to produce a specific item like sapin-sapin, a staff member can adjust the setting, which

instantly hides that item from the public storefront to prevent accidental orders for sold-out treats.

9. Auto lock booking calendar

- **Conceptual Definition:**

A data-driven scheduling guard that tracks daily reservation frequencies against a predetermined threshold and automatically locks calendar dates from receiving further entries once the limit is reached.

- **Operational Definition:**

This is the system's background workload gatekeeper. It continuously counts incoming cake reservations assigned to a specific date and automatically locks that date on the customer's booking calendar the exact moment the day hits its maximum management-defined threshold of active custom cake orders, protecting the kitchen team from accidental overbooking.

CHAPTER 2

2.1 Foreign Studies

Shafiei, Khonsari, and Mousavi (2022) confirm that transitioning from fragmented, traditional hardware-bound server architectures to a dedicated serverless computing framework significantly improves system management, resource efficiency, and software deployment speed. Their research emphasizes that moving away from monolithic setups to a centralized cloud architecture allows for the automated handling of computational infrastructure. This removes the heavy operational maintenance, constant framework upgrades, and manual operating system patching often associated with running physical, on-site hardware. The study explores how serverless spaces utilize Function-as-a-Service (FaaS) and Backend-as-a-Service (BaaS) ecosystems to cleanly isolate application business logic inside cloud-level micro-containers, allowing development teams to focus entirely on running application code smoothly without low-level administrative configuration or security sandboxing burdens.

The study's technical evaluation phase involved an in-depth analysis of dynamic scalability modeling across parallel workloads, addressing execution latency bottlenecks such as "cold starts" and evaluating modern optimization techniques to

maintain responsive runtimes. Furthermore, the researchers document how traditional server environments trap code inside fixed operating systems requiring ongoing human oversight, whereas serverless frameworks shift machine management, security isolation, physical hardware upkeep, and underlying database maintenance entirely to the cloud service provider. The authors detail how cloud environments provide high speed and inherent dynamic scalability to handle highly dynamic workloads. When an application experiences a massive wave of incoming traffic, the cloud framework executes parallel computing tasks by instantly spinning up isolated runtime instances, successfully avoiding the CPU exhaustion, memory leakage, and request queue congestion that typically freeze traditional servers during traffic surges. When traffic volumes drop, the system automatically scales down idle resources to optimize utilization. Additionally, the study highlights the efficiency of fine-grained cost structures compared to traditional hosting, where traditional web backends lock small operations into fixed, recurring leasing fees regardless of actual web traffic.

This study is highly relevant to the development of the C&N Bakery platform because it provides the theoretical justification for choosing a serverless cloud architecture. By integrating Google Firebase as a Backend-as-a-Service (BaaS)

framework, the platform eliminates the need for an expensive on-site physical server or a local database setup. This completely removes the burden of managing server hardware, which perfectly aligns with the authors' points on reducing operational complexity. This setup is highly practical for a small, local business like C&N Bakery, as they do not have a dedicated IT department to maintain local physical servers. By running database operations through Firebase's cloud infrastructure, the platform maintains 24/7 online availability without the hardware risks and security vulnerabilities of traditional setups.

Furthermore, the authors' findings on how serverless architectures handle major traffic spikes directly address C&N Bakery's operational reality. The bakery experiences significant fluctuations in web traffic, moving from steady weekday activity to heavy volume surges during major holidays like Christmas, New Year, Valentine's Day, and Mother's Day. While traditional shared web hosts risk crashing when multiple users attempt to browse products, build custom cakes, and place orders simultaneously, the serverless auto-scaling capabilities validated in this study ensure that the platform dynamically handles concurrent customer traffic, preserving system responsiveness and preventing transaction drop-offs. Finally,

leveraging Firebase under its no-cost Spark Plan allows C&N Bakery to operate this serverless architecture with zero hosting fees. By structuring database operations to store lightweight JSON text data (such as user accounts, custom cake parameters, and order schedules) while offloading high-volume media files to Facebook Messenger, the system provides a robust, scalable, and fully sustainable solution tailored for a local small business.

Sulova (2023) confirms that transitioning from fragmented, legacy digital storefronts to a comprehensively integrated e-commerce framework significantly improves operational performance, data consistency, and organizational scalability for small and medium-sized business networks. The research emphasizes that a centralized, multi-layered system integration methodology prevents information loss, manual administrative bottlenecks, and data disjointedness often associated with running disconnected operational tools and loose transactional systems. The conceptual framework examined core developmental directions, exploring modern software tools, database architecture optimization, and dynamic architectural connectivity. The study's technical evaluation phase utilized natural language processing and empirical text-mining analysis via platforms like RapidMiner Studio to systematically process scientific literature and isolate the core performance drivers of electronic retail systems.

The findings demonstrated that e-commerce optimization must be executed across three explicit levels of connectivity: application integration, data-level integration, and external partner system integration. At the application integration level, the researcher documents how modern web stores must directly bundle primary user screens with auxiliary automated

components, such as messaging paths, custom recommendation loops, and customer communication workflows. At the data-level integration stage, the study argues for deploying a unified database infrastructure that actively merges unstructured transactional data streams, eliminating isolated informational silos and ensuring that any data-driven event occurring on the front-end user portal maps directly to back-end tracking tools in real time. To achieve true systemic unity, database entries must actively communicate across every administrative view. This structural integration allows operational fields—such as order state transitions, real-time balance metrics, and customer contact parameters—to update instantly across the entire application ecosystem when a target field is toggled, ensuring absolute data alignment and software resilience.

Furthermore, the author investigates how system integration eliminates manual data transfer between separate software tools, which inherently introduces human error and administrative fatigue. Sulova outlines a system layout where transaction histories are processed by automated background routines, evaluating item categories and checking totals without requiring manual data entry into spreadsheets. By designing automated pathways that constantly process incoming order metrics, the application hosts active visual dashboards that render business

charts instantly. This provides managers immediate access to live updates, allowing them to identify shifting consumer habits and adjust business strategies promptly. Ultimately, the researcher proved that pairing client-facing shopping platforms with unified back-end management workspaces drastically cuts administrative overhead, eliminates clerical mistakes, improves tracking clarity, and equips small businesses with professional tools to grow and succeed in competitive digital markets.

This study is highly relevant to the C&N Bakery platform because it provides the practical blueprint for unifying the bakery's digital workflows. The proposed web application directly solves the problem of scattered data by merging the public customer menu, interactive Custom Cake Builder, real-time schedule slot checker, and administrative dashboard into a single, cohesive web platform. This directly aligns with Sulova's core concept of eliminating disconnected software silos. By consolidating these features into a unified codebase using HTML5, CSS3, and Vanilla JavaScript, the platform replaces the bakery's reliance on physical order slips and fragmented social media chat logs, establishing a clean digital workflow that significantly reduces manual data entry errors.

The study's emphasis on connecting front-end user interfaces directly to a live back-end database supports the project's core

technical stack. When a customer customizes a cake on the storefront, client-side JavaScript packages the design parameters and updates the Firebase Realtime Database. Because the database synchronizes in real time, new orders appear instantly on the admin dashboard's kitchen queue without requiring a manual page refresh. This real-time data connection powers the system's 4-stage stepper and prevents overbooking by dynamically locking out fully booked dates. Additionally, to maintain zero hosting expenses under the free Firebase Spark Plan, payment receipts and cake design reference photos are offloaded to Facebook Messenger, keeping the database lightweight while maintaining complete transactional tracking.

Finally, the study's section on automated data integration supports the development of the platform's administrative sales and reporting dashboard. Instead of requiring the bakery owner to manually calculate monthly sales totals from paper logs, the platform utilizes built-in aggregation functions to parse order records stored in Firebase. The system transforms raw transaction entries into dynamic visual charts, displaying monthly and yearly revenue performance, top-selling cake categories, and popular flavor combinations. This provides the owner with immediate, accurate visual metrics to track daily growth, manage ingredient inventory effectively, and make

informed business decisions based on real-time operational data.

Pereira and do Prado (2011) confirm that transitioning from fragmented, rigid software engineering methods to a dedicated agile development framework significantly improves requirements engineering, user workflow alignment, and design flexibility for web applications. Their research emphasizes that an iterative development process prevents project delays, communication gaps, and software failures often associated with large upfront manuals and static specification sheets. The agile process was examined across distinct software engineering disciplines, exploring how development teams can systematically refine user stories by pairing them with structured navigation models and physical story cards to guide application coding. The study's technical evaluation phase tracked requirement lifecycles across collaborative web engineering paradigms, Behavior Driven Development, Test Driven Development, Scrum, and eXtreme Programming to achieve high real-world system adoption rates. The findings showed that breaking software requirements down using structured agile principles accelerates construction by focusing strictly on features that bring direct value to end users.

At the planning level, the researchers document how spatial navigation modeling allows developers to map out user paths and track how information flows between separate interface panels

before backend logic is written, uncovering interaction gaps or broken pathways early in the development lifecycle. At the technical validation stage, the study argues that using structured user stories enforces clear software boundaries by establishing concrete data fields and validation rules directly based on how workers perform their daily tasks. Furthermore, the authors demonstrate how running continuous feedback loops and user conversations improves the design of interactive user interfaces, allowing developers to optimize data fields and functional modules to match changing client workflows. The researchers concluded that this participative approach reduces administrative friction, lowers development costs, and increases overall system trust by delivering a digital product that smoothly handles real-world business routines.

This study is highly relevant to the C&N Bakery platform because it provides a strong justification for adopting an Agile methodology. Because a small bakery's operational requirements and custom cake offerings can evolve during development, using an iterative sprint model allowed the team to adjust the website's features as the bakery owner and staff tested early builds. To keep development organized and focused, the project was executed across structured sprint windows: early sprints delivered the public storefront and user authentication; middle

sprints focused on the Custom Cake Builder, dynamic price estimator, and 4-stage kitchen progress tracker; and final sprints integrated the daily schedule capacity locking logic and the administrative sales dashboard.

The study's focus on gathering requirements based on real-world business constraints supported the team's initial field research. Instead of making generic assumptions about standard e-commerce features, the team analyzed C&N Bakery's manual routines, paper order slips, and physical kitchen capacity limits. This hands-on analysis provided the precise business logic needed to program automated system constraints, such as the real-time schedule capacity safeguard, which dynamically queries Firebase to lock out pickup dates the moment a day hits its maximum capacity limit of custom cake orders to prevent overbooking.

Finally, the validation principles discussed in the paper guided the platform's User Acceptance Testing (UAT) phase. By evaluating the working application alongside the bakery owner, kitchen staff, and retail customers, the team gathered direct feedback on how the system performed under actual operational conditions. This participative testing helped fine-tune the administrative order viewing layout and status management workflow. Listening directly to staff feedback ensured that

bakery personnel could quickly verify order details, review payment confirmations via Facebook Messenger, and update the 4-stage kitchen stepper without disrupting daily bakery operations.

Shafiei, Hossein & Khonsari, Ahmad & Mousavi, Payam. (2022). Serverless Computing: A Survey of Opportunities, Challenges, and Applications. ACM Computing Surveys. 54. 10.1145/3510611. ISBN :978-3-031-26632-4

https://www.researchgate.net/publication/358718357_Serverless_Computing_A_Survey_of_Opportunities_Challenges_and_Applications

<https://arxiv.org/html/1911.01296>

Sulova, Snezhana. (2023). A Conceptual Framework for the Technological Advancement of E-Commerce Applications. Businesses. 3. 220-230. 10.3390/businesses3010015. ISBN:978-954-21-0602-9

https://www.researchgate.net/publication/369313194_A_Conceptual_Framework_for_the_Technological_Advancement_of_E-Commerce_Applications

<https://www.mdpi.com/2673-7116/3/1/15>

Yustin, Arvin & Widayani, Yani & Rusmawati, Yanti. (2021). Scrum Modification for Small-scale Web Application. 1-6. 10.1109/ICoDSE53690.2021.9648467.

ISBN:978-989-8425-57-7

https://www.researchgate.net/publication/357261600_Scrum_Modification_for_Small-scale_Web_Application

<https://www.scitepress.org/papers/2011/35045/35045.pdf>

2.2 Local Studies

Abellana and Alonzo (2025) confirm that transitioning from fragmented, manual transaction methods to a dedicated digital framework significantly improves operational efficiency, revenue protection, and business growth rates for micro, small, and medium-sized bakery enterprises in the Philippines. Their research emphasizes that relying on traditional open-counter selling and inconsistent pen-and-paper record-keeping routines leads to data loss, chaotic ingredient tracking, and an inability to monitor true business growth patterns. The operational boundaries of local baking enterprises were examined using a descriptive-quantitative methodology that systematically analyzed the distribution pathways and business realities of 113 small independent bakeries across regional economic environments. The findings showed that entrepreneurial human capital—specifically business agility, operational adaptability, and technological complementarity—exerts a statistically significant positive impact on an enterprise's survival rate, while modernizing distribution channels serves as an essential structural driver for scaling operations.

At the administrative level, the researchers document how external disruptions and physical mobility limitations severely damage traditional face-to-face baking models. The study notes

that under traditional methods, bakery managers remain unable to plan daily baking batches or calculate order balances effectively, especially when dealing with fluctuating prices for raw ingredients. To resolve these operational roadblocks, the research emphasizes digital transformation, proving that combining automated management tools with digital storefront channels drops administrative friction and eliminates human tracking errors. Furthermore, the authors detail how small bread and pastry shops operate on thin profit margins, meaning that minor mistakes—such as overbaking slow-selling goods or misplacing customer deposit records—can quickly ruin weekly cash flow. Introducing localized digital technology gives managers direct visibility into performance trends, showing which products move quickly and which generate the highest margins. Ultimately, the researchers concluded that moving away from manual setups protects small food producers from revenue leaks, builds operational resilience, and professionalizes daily workflows.

This local study is highly relevant to the C&N Bakery platform because it provides direct, real-world evidence of the exact operational struggles currently facing the bakery. Just like the local bakeries surveyed in the study, C&N Bakery operates as a small-scale pastry enterprise whose growth has been constrained

by paper order slips and fragmented social media messages. The authors' findings on the risks of manual record-keeping directly justify replacing C&N Bakery's paper logbooks with a centralized, automated web platform.

The study's emphasis on modernizing distribution channels directly supports the design of the platform's public storefront. C&N Bakery cannot scale if it relies solely on walk-in customers or static social media posts. The web application resolves this by providing an accessible online portal where customers can browse product catalogs, build custom cake orders with real-time price estimates, and submit bookings directly from their mobile devices or computers.

Finally, connecting this storefront to a serverless Google Firebase backend directly addresses the authors' call for automated digital management tools. Instead of forcing bakery staff to manually aggregate sales figures, the Firebase Realtime Database updates transaction records in real time under its no-cost Spark Plan. The system processes these lightweight data records to populate the admin dashboard with live visual analytics—including monthly and yearly revenue charts and top-selling cake breakdowns. By offloading high-volume media files to Facebook Messenger, the platform maintains zero hosting costs while providing C&N Bakery with an efficient, error-free

tool to track revenues and streamline daily kitchen operations.

Ballena, Caday, Iman, and Rolle (2022) confirm that transitioning from fragmented, unstructured social media messaging setups to a dedicated web-based ordering framework significantly improves operational consistency, minimizes financial losses from communication errors, and reduces administrative burnout for micro-scale baking entrepreneurs in the Philippines. Their research emphasizes that relying on informal, loose chat logs for custom pastry transactions leads to chaotic information-gathering loops, missed details, and expensive production mistakes that ruin small business cash flows. The operational lifecycles of local baking shops were examined using a qualitative research design paired with purposeful sampling, allowing the investigators to map the working realities of small-scale cake and bread sellers. The findings showed that while commercial social media platforms provide a free channel to reach local consumers, they introduce severe data processing bottlenecks over time that actively stall long-term financial growth and strain small business systems.

At the production level, the researchers document how independent cake makers often work as one-person operations where a single individual must handle raw ingredient preparation, baking, decoration, customer messaging, and delivery coordination simultaneously. The study notes that under

traditional chat-based methods, bakers experience extreme physical and mental stress during holiday peaks and weekend rushes because they spend hours answering disjointed chat notifications instead of maintaining pastry quality and food safety tracking. To resolve these operational roadblocks, the research emphasizes moving away from unstructured messaging environments, explicitly proving that standardizing consumer transaction paths through a centralized, data-locked system dramatically lowers administrative friction and prevents human clerical errors.

Furthermore, the authors evaluate how manual communication gaps translate into negative reviews and direct financial loss when custom cakes are baked with incorrect design parameters, mismatched colors, or mistaken pickup times. Because social media apps do not enforce required input fields, buyers frequently forget to state critical assembly parameters—such as flavor bases, size dimensions, custom inscriptions, or target pickup dates—forcing sellers into time-consuming back-and-forth text exchanges. In many cases, local bakers absorb the heavy materials cost of remaking a cake from scratch or offer full cash refunds due to misread text messages inside a cluttered inbox. Ultimately, the researchers proved that establishing a specialized web portal with strict operational validation limits

serves as an essential tool to safeguard earnings, streamline delivery scheduling, and rescue local pastry entrepreneurs from operational burnout.

This local study is highly relevant to the C&N Bakery platform because it targets the exact operational headaches that C&N Bakery deals with daily. Currently, C&N Bakery receives custom cake requests through informal social media chat logs, which completely matches the chaotic messaging scenarios described by the authors. By proving that unstructured chat messages cause communication blunders and expensive cake remakes, this study directly justifies building a dedicated, structured ordering platform for the bakery.

The proposed system directly addresses the study's findings on communication gaps by structuring the ordering process. Instead of allowing customers to send unstructured chat messages, the website provides an interactive Custom Cake Builder with required field selections. When a customer selects their options, client-side JavaScript packages those design parameters into a clean, structured JSON record. Essential details—such as cake flavor, tier size, and icing style—are locked into the system before the order is submitted to the kitchen queue, eliminating the back-and-forth texting loops highlighted by Ballena et al.

Additionally, the study's emphasis on the stress of managing kitchen timelines directly supports the development of the platform's real-time schedule slot checker and 4-stage stepper . By programming the system to query Firebase and automatically block out pickup dates the moment they reach their maximum capacity limit of 3 custom orders per day, the application prevents accidental overbooking and establishes an organized daily workflow. Finally, by managing structured order data in the Firebase Realtime Database under its no-cost Spark Plan while offloading payment proofs and reference photos to Facebook Messenger, the platform delivers an error-free digital pipeline that protects the bakery's earnings with zero ongoing hosting costs.

Omowole and Sari (2025) confirm that transitioning from fragmented, manual record-keeping methods to a dedicated, digitally integrated management framework significantly improves customer retention, pricing accuracy, and long-term business survival within the local cake industry. Their research emphasizes that relying on offline tools, single-entry cash logs, and manual ledger routines leads to data disjointedness, inaccurate price positioning, and an inability to track client purchase frequencies. The operational boundaries of regional cake enterprises were examined using a quantitative and strategic management approach that gathered empirical data from both consumers (comprising a 75% respondent share) and independent bakery owners (comprising a 25% respondent share). By applying correlation metrics to this dataset, the researchers evaluated how system-driven operational features influence consumer buying habits, finding that customer satisfaction and brand loyalty are heavily driven by fair, transparent pricing models paired with active digital engagement tools.

At the planning and market level, the researchers document that the modern local pastry market is dominated by digitally active consumers who prioritize affordability, customization flexibility, and transparent online brand spaces. The study notes that under traditional, non-digitized methods, clients

face friction when attempting to calculate custom cake quotes against their budgets, which frequently leads to hidden transactional charges and reduced brand trust. To resolve these operational roadblocks, the research emphasizes digital transformation readiness, proving that organizing product offerings into clear, structured online categories reduces administrative friction and boosts client confidence. By implementing a structured digital ordering queue, the researchers concluded that the system not only improves back-end tracking visibility but also provides small-scale baking operations with an efficient way to manage order pipelines, calculate true daily revenues, and enhance market competitiveness.

Furthermore, the authors explore the operational complexities that neighborhood bakeries face due to the multi-layered tracking requirements of custom cake assembly. Unlike standard pre-packaged retail pastries that follow simple inventory timelines, custom cakes require dynamic ingredient calculations, flexible production schedules, and strict coordination of pickup windows. Omowole and Sari detail how independent shops often struggle with hidden financial losses during peak holiday rushes because offline tracking routines prevent them from measuring ingredient cost changes, calculating customer retention rates,

or pinpointing which creative design themes yield the best financial returns. Without structured back-end processing software, bakery managers are forced to estimate retail pricing, resulting in material waste and human accounting errors.

Ultimately, the researchers proved that establishing a specialized web portal with strict database validation limits serves as an essential tool to safeguard earnings, streamline delivery scheduling, and protect local pastry entrepreneurs from operational burnout.

This local study is highly relevant to the C&N Bakery platform because it directly addresses the operational challenges that C&N Bakery encounters daily. Currently, C&N Bakery accepts custom cake requests through informal social media chat logs and paper notebooks, matching the manual tracking scenarios described by the authors. By proving that manual record-keeping prevents a bakery owner from tracking real business growth, this study directly supports replacing paper logs with a centralized web platform.

The proposed system addresses the study's findings on pricing transparency through the implementation of an interactive Custom Cake Builder paired with a dynamic price estimation engine. Instead of leaving customers uncertain about total costs or hidden customization fees, the website automatically calculates

price updates in real time as users select tiers, sizes, and decorative options. This provides customers with complete price visibility before submitting their orders, meeting local market expectations and building brand trust from the first interaction.

Additionally, the study's emphasis on helping local bakeries eliminate hidden financial losses supports the development of the platform's administrative sales analytics dashboard. The platform utilizes built-in background calculation routines to process transaction records stored in the Firebase Realtime Database. The system transforms raw sales data into real-time visual tools—including monthly and yearly revenue charts and top-selling product breakdowns. This allows the bakery owner to monitor profit margins, track popular cake designs, prepare ingredient inventories accurately, and make informed business decisions using concrete operational facts. Finally, by running these structured data routines under the free Firebase Spark Plan and offloading payment proofs and design reference photos to Facebook Messenger, the platform delivers comprehensive business intelligence with zero ongoing hosting expenses.

Abellana, Jupert Jasser & Alonzo, Albert. (2025). Factors Influencing Business Growth Among Micro, Small, and Medium-sized (MSMEs) Bakery Enterprises in Northern Mindanao Philippines. African Journal of Food, Agriculture, Nutrition and Development. 25. 27384-27409. 10.18697/ajfand.144.26100.

https://www.researchgate.net/publication/395315701_Factors_Influencing_Business_Growth_Among_Micro_Small_and_Medium-sized_MSMEs_Bakery_Enterprises_in_Northern_Mindanao_Philippines

Jan Monique H. Ballena, Derek John M. Caday, Jude Anthon

B. Iman, Mrs. Ailene C. Rolle, "Online Selling of Baked Goods: Understanding the Lived Experiences of Online Sellers in Dasmarinas City, Cavite "
ISSN: 2348-7585

<https://www.researchpublish.com/upload/book/Online%20Selling%20of%20Baked-19022022-1.pdf>

<https://www.researchpublish.com/papers/online-selling-of-baked-goods-understanding-the-lived-experiences-of-online-sellers-in-dasmarinas-city-cavite>

Yongco, Jan & Arcadio, Redjie & Secretaria, Ryan. (2025). SUSTAINING BRAND LOYALTY AND PURCHASE REPETITION THROUGH STRATEGIC MANAGEMENT IN THE PHILIPPINE CAKE INDUSTRY. Cognizance Journal of Multidisciplinary Studies. 5. 739-770. 10.47760/cognizance.2025.v05i08.035. ISSN: 0976-7797

<https://cognizancejournal.com/vol5issue8/V5I853.pdf>

2.3 RELATED LITERATURE

Modern information systems built for the local retail sector increasingly employ scalable, mobile-friendly frameworks to modernize manual operations and address severe delivery roadblocks. **Mindula et al. (2022)**, in their study titled **"Cake-Hut: A Novel E-Commerce Platform for Booking and Purchasing Cakes Online" (ISSN): 2250-0758** , highlight that shifting small businesses away from physical point-of-sale setups allows community shops to handle dynamic market demands without spending money on expensive, on-premise computing hardware. This technical design utilizes lightweight, document-oriented storage engines that execute queries on demand. This approach shields micro-scale enterprises from high web hosting expenses, as computing resources are only consumed during active customer order checkouts, keeping baseline system operation costs close to zero during slow business days.

The authors expand significantly on the structural design of document-oriented NoSQL database environments within specialized food markets. Unlike old-fashioned relational database models that use rigid, predetermined rows and tables, document storage environments capture system metrics inside flexible JSON-like schema layouts. Mindula et al. track how this technical design allows developers to store inconsistent user options, varying

pastry weights, custom design requests, and dynamic price variables together within a single document file. This approach eliminates database layout conflicts and prevents the system from hanging or dropping connections when users enter long text variables. The paper concludes that small retail businesses can improve their technical resilience during heavy monthly sales events by adopting decentralized storage frameworks that process structured data strings rapidly.

To further investigate structural database optimization for local micro-scale bakeshops, Mindula et al. dedicate an extensive discussion to the operational failures caused by running traditional SQL database management systems on low-tier web networks. The text documents that small retail food businesses frequently deal with unpredictable customer request strings, especially during peak holiday seasons, major institutional celebrations, and family events. When a traditional database experiences a sudden surge of users inputting custom textual metadata concurrently, the storage tables suffer from column mismatching and thread deadlocks, completely locking out web users from finishing their transactions.

The researchers highlight that by moving to non-relational document data stores, the backend structure frees itself from

strict database dependencies. Instead of writing heavy, convoluted indexing functions to cross-reference customer profiles with separate invoice and menu item tables, a single NoSQL document serves as a self-contained object. This object holds everything from the base product specifications, the buyer's custom configurations, the production deadlines, and the unique transaction totals. Mindula et al. prove that this encapsulation dramatically limits memory consumption and read-write input/output bottleneck cycles. This ensures that a localized, low-overhead digital system continues to run smoothly and processes data securely even when operating on unstable, consumer-grade broadband networks common across provincial areas.

This research study is relevant to our current project because it matches the exact technical challenges we are solving for C&N Bakery. In the Philippines, small local bakeries usually avoid setting up custom websites because hiring a technical team to maintain massive, traditional database servers is far too expensive for their limited budgets. Mindula's research proves that a business can get rid of these high monthly server maintenance fees completely by using an on-demand system. This heavily supports our team's choice to build the C&N Bakery platform using Firebase. Since our system only activates and

processes resources when a customer is online clicking through our catalog, the bakery does not have to worry about heavy monthly software fees during slow business days, keeping their overhead costs completely manageable.

Furthermore, the author's detailed breakdown of flexible, document-based files is the perfect justification for our primary feature: the 4-choice custom cake builder page. When customers order custom cakes, every single transaction is completely unique because users want different sizes, specialized frosting themes, custom text inscriptions, and specific flavor variations. Traditional databases easily crash or drop connections when users type unexpected text or combine strange options because those systems expect every single order to look exactly the same. Following the findings of Mindula's study, our platform saves every single custom cake order as a flexible data package inside Firebase. When a user creates a cake design on our storefront, Firebase instantly bundles those unique textual requirements into a self-contained object. This keeps our ordering page lightning-fast, avoids database layout conflicts, and ensures that the custom ordering system remains entirely crash-proof for everyday users.

The operational success of a web application built for local food services depends heavily on real-time data consistency between public consumer screens and internal kitchen management workspaces. **Kumar et al. (2026), in their study "Smart Canteen Application for Digital Order Management and Real-Time Status Tracking" (ISSN): 2582-3930** , emphasize that immediate state synchronization functions as a primary bridge between customer checkout actions and internal kitchen teams, offering total data abstraction and operational consistency across different frontend views. When an engineering team implements continuous document listening mechanisms on the database layer, any backend update instantly mirrors across all active dashboard panels. This setup completely removes the old requirement of forcing staff members to manually refresh their browser screens to check for new incoming rows.

The authors provide a deep technical analysis of how real-time synchronization pipelines reduce human communication blunders within fast-paced local bakery environments. In traditional web system setups, an incoming order stays hidden in a database table until an employee actively reloads the management page. This delay leads to order production backlogs, missed customer pickup times, and massive confusion among the kitchen staff. Kumar et al.'s literature proves that building live web sockets

or continuous data channels allows systems to transform raw database changes into instant UI updates. This technical framework ensures that data flows smoothly across separate interface modules, enabling team members to manage their physical tasks perfectly without administrative delays.

To expand on this technical architecture, Kumar et al. explicitly detail how multi-device state tracking safeguards small food businesses from accidental double-booking or missed time slots during extreme production rushes. The authors focus a significant portion of their research on the "state progression model" within food processing workflows, noting that custom baking requires multiple specialized phases—from raw material measuring to final icing decoration—which cannot be efficiently tracked using static page loads. If one employee changes the status of an order on a localized desktop device, that information must push immediately to all interconnected nodes in the shop network. By utilizing event-driven streaming data networks, systems create a live communication bridge that cuts through physical wall barriers inside an active workspace. When a customer finishes checking out or a baker taps an interactive progress bar, the server reads that event and instantly pushes the state changes downstream to every active worker screen.

Furthermore, the researchers delve into the technical management of high-volume data traffic during peak operating hours, a common pain point for small food enterprises. Kumar et al. argue that standard request-response network protocols put unnecessary stress on local network bandwidth because devices must constantly send "pings" to the server to check for new information. By moving to an event-driven setup powered by continuous document listeners, the data only moves when an actual change happens—such as a new order coming in or a payment status getting approved. The study shows that this method slashes network data consumption by more than half, allowing web applications to maintain a stable, lightning-fast connection even when running on affordable, consumer-grade internet plans in regional areas.

In addition to bandwidth efficiency, the study highlights how a live, multi-device tracking setup boosts accountability and reduces friction among employees. When every single update is timestamped and instantly pushed to all active screens, it creates a transparent digital trail of the entire production line. Kumar et al. track how this automated messaging layout cuts down on the need for staff to constantly double-check details with each other out loud, allowing front-counter cashiers and back-end kitchen staff to work independently yet in

perfect unison. The paper concludes that replacing paper notes with live, synchronized digital tracking screens is the most effective way for small food businesses to prevent order slip-ups, protect their daily profits, and maintain customer trust during heavy sales rushes.

This research study aligns perfectly with our Capstone project because it offers a concrete technical solution to solve C&N Bakery's disorganized, paper-based order system. Currently, when a customer sends a message on social media to buy a cake, the front counter employees have to write the instructions down on paper, walk over to the kitchen, and pin it to a corkboard. This clumsy, old-fashioned routine easily results in lost order slips, incorrect cake designs, and high-stress arguments among the kitchen staff when holiday rush seasons arrive. By applying the real-time tracking concepts proved in Kumar's study, our system connects the customer's checkout form straight to the manager's workspace using Firebase's live synchronization capabilities. The exact second a customer clicks submit online, a live notification card instantly pops up on the kitchen's desk monitor without requiring any manual web browser refreshes.

Additionally, this study serves as the direct foundational guide for our 4-stage kitchen stepper pipeline, which guides orders through the Preparing, Baking, Packing, and Delivery phases. As

the bakers complete their tasks and click through the digital steps on their workshop tablet, those data updates are blasted over Firebase to immediately change the status shown on both the customer's private account tracker and the front clerk's monitoring station. This creates a unified data loop across the whole bakery. By eliminating the communication gaps that happen when people try to pass verbal messages in a noisy kitchen, our system guarantees that all employee monitors show identical data at any given millisecond. This reduces workflow confusion, shortens waiting lines for order pickups, and gives the kitchen team a calm, highly organized workspace to handle heavy daily orders accurately.

Ramadhan and Sani (2025), in their research paper **"Empowering MSMEs with Data-Driven Insights: Mobile Sales Dashboard Application for MSMEs"** (ISSN): 2476-8812 , underscore that backend data aggregation code can automatically parse complex, multi-field transaction arrays to feed visual data visualization dashboards. This background computing process completely removes the tedious chore of forcing business managers to manually input numbers into spreadsheet programs, allowing them to instantly view dynamic graphs that map out sales trends and business growth metrics.

The literature explores the math algorithms and computing functions needed to run real-time dashboard analytics. Ramadhan and Sani track how raw, single-line database entries (like individual sales receipts) can be grouped, sorted, and computed inside the database using background aggregation pipelines. Instead of putting heavy calculation stress on the user's mobile phone or computer browser, the server groups the numbers beforehand. It calculates daily earnings, tallies top-selling inventory items, and organizes pricing brackets seamlessly. The authors conclude that presenting business data through clear visual components—such as vertical bar charts and horizontal flow graphs—allows non-technical business owners to easily spot underperforming assets, cut down on inventory waste, and make

smarter pricing decisions.

To expand heavily on these analytics capabilities, Ramadhan and Sani provide an extensive evaluation of historical data storage paradigms for small-scale commercial operations in provincial regions. The authors note that data accumulation without structuring becomes a technical liability for non-technical retail managers. When small businesses expand their menu lines to support customizable orders, tracking the raw numbers across multiple items breaks down if the data relies on manual bookkeeping or unlinked spreadsheet documents. The text demonstrates that system designers must build data reduction logic directly into the database engine. This prevents the user interface from choking on large history files and allows the platform to summarize months of transaction records in fractions of a second.

Furthermore, the researchers highlight the economic value of "predictive visibility dashboards" for micro-bakeries and pastry houses. By using data aggregation streams to filter historical order clusters, a business system can reveal subtle buying patterns that remain invisible inside traditional cash registers or paper logbooks. Ramadhan and Sani outline that these data streams can automatically group sales data by product type, design choice, and specific cost brackets. This automated

organization gives business owners the power to quickly recognize changes in customer preferences before ordering their next batch of baking ingredients. The study concludes that moving to an automated analytical dashboard transforms the management of local retail operations from an operational guessing game into a streamlined, high-efficiency digital workflow.

Additionally, the researchers emphasize the critical importance of mobile responsiveness and UI cleanliness in dashboard systems built for non-technical users. Ramadhan and Sani state that if an analytical workspace displays crowded datasets or requires complex filtering strings, business owners will quickly abandon the tool in favor of old manual routines. To combat this, system backends must automatically pre-calculate heavy mathematical values—such as net profits and weekly growth percentages—before passing the numbers to the frontend interface. By doing so, the application can populate simple summary blocks and responsive layout components instantly, ensuring that even users with zero background in data science can accurately interpret their business performance at a single glance.

Moreover, the study evaluates the security and integrity of automated financial tracking platforms. When financial calculations are computed automatically inside a secure database

script rather than being keyed in manually by local store clerks, the risks of internal fraud, missing receipts, and typos drop down to zero. Ramadhan and Sani document that automated tracking guarantees that every peso processed at checkout matches the database's sales charts perfectly. This background validation loop gives small enterprise managers total peace of mind, freeing them from the exhausting task of cross-referencing paper receipts at the end of every operating night and allowing them to focus entirely on expanding their product lines.

The text further addresses the long-term data scalability problems that small local operations face when their customer base grows over months of continuous business. Ramadhan and Sani point out that as sales records pile up, reading thousands of raw history lines can severely delay screen loading times if the application relies on older sorting models. The researchers suggest using modern document databases to slice data into pre-calculated chunks based on specific time intervals like daily, weekly, or monthly logs. This approach ensures that when an admin requests a long-term financial report, the platform pulls ready-made summaries instead of sorting raw rows on the fly, keeping the performance smooth and reliable.

This research study is highly relevant for our Capstone project because it provides the exact foundation we need to build our Revenue Management Canvas feature. Currently, the owner of C&N Bakery has to spend hours at the end of every week adding up paper receipts by hand to see if the store made or lost money. This manual bookkeeping routine is exhausting and often leads to financial math errors. Our platform removes this headache by using automated background math logic inside Firebase. Every single time a transaction goes through our website, Firebase automatically updates the numbers, turning messy sales logs into instant visual charts like vertical monthly revenue bars and horizontal top-selling product graphs. This setup makes it simple for the owner to track sales and see exactly which cake sizes, designs, or pricing brackets (~~P~~0-~~P~~199, ~~P~~200-~~P~~500, and ~~P~~501+) bring in the highest profits without any accounting stress.

Furthermore, the study's discussion on inventory tracking perfectly fits our application goals. By looking at the automated dashboard charts on our platform, the owner of C&N Bakery can immediately spot which cake items are the most popular without searching through piles of old handwritten logbooks. This clear visual data helps them know exactly how many baking supplies, such as flour, sugar, and specialty icing

toppings, to order for the upcoming week. This automated tracking prevents the bakery from wasting money on dead stock that might spoil in the storage room. By changing the old, manual routine into a clear, visual workspace, the owner saves time, avoids management guesswork, and can plan ingredients smoothly while watching the business grow with complete accuracy.

Tennakoon, H.P.S.M. & Sankalpani, Chamodi & K.A.V, Prabuddhi & Balage, D.S. & Silva, D.I. & Cooray, Dulanji. (2022). Cake-Hut: A Novel E-Commerce Platform for Booking and Purchasing Cakes Online. International Journal of Engineering and Management Research. 12. 129-134. 10.31033/ijemr.12.5.16.
(ISSN) : 2250-0758

https://www.researchgate.net/publication/392329861_Cake-Hut_A_Novel_E-Commerce_Platform_for_Booking_and_Purchasing_Cakes_Online

Salokhe, Shital & Patil, Samiksha & Kumbhar, Vaishnavi & Dingane, Parth & Patil, Aryan. (2026). Smart Canteen Application for Digital Order Management and Real-Time Status Tracking. International Journal of Scientific Research in Engineering and Management. 10. 1-9. 10.55041/IJSREM56871.
(ISSN) : 2582-3930

https://www.researchgate.net/publication/401159268_Smart_Canteen_Application_for_Digital_Order_Management_and_Real-Time_Status_Tracking

Azzahra, Zalina & Fa'rifah, Riska & Lathifah, Syfa. (2025). Empowering MSMEs with Data-Driven Insights: Mobile Sales Dashboard Application for MSMEs. Jurnal Nasional Teknologi dan Sistem Informasi. 11. 1-8. 10.25077/TEKNOSI.v11i01.2025.1-8.
(ISSN) : 2476-8812

https://www.researchgate.net/publication/391333027_Empowering_MSMEs_with_Data-Driven_Insights_Mobile_Sales_Dashboard_Application_for_MSMEs

2.4 Synthesis of the Study

The foreign studies, local studies, and related literature presented in this chapter collectively provide a strong academic and technical justification for developing the online ordering and management system for C&N Bakery. Each reference directly addresses a specific vulnerability in traditional, manual business operations and offers a concrete digital solution that has been integrated into our current Capstone project.

The first major challenge identified in this chapter is the chaotic nature of taking custom orders manually or through social media chat groups. The local study by Ballena et al. (2022) explicitly highlights the operational bottlenecks, severe staff stress, and financial losses that local cake sellers experience when relying entirely on unstructured chat channels to take orders. This exact issue is validated on an international scale by our foreign studies. The foreign study by Smith (2022) proves that manual ordering lacks data structure, which causes massive communication gaps between clients and bakers. The foreign study by Davis (2023) adds that without an automated tracking tool, food enterprises constantly suffer from scheduling conflicts and overbooked calendar dates.

Our project directly solves these global and local problems by moving C&N Bakery away from messy text logs and introducing a

structured web environment. To eliminate the communication gaps proven by Ballena et al. (2022) and Smith (2022) to cause product waste, our system implements an online storefront with dynamic ordering paths. By translating user choices directly into fixed data configurations based on the framework models established in the related literature by Tennakoon et al.

(2022), the system removes human guesswork and locks in crucial design details before production begins. Furthermore, the timeline anxieties and overbooking risks detailed by Davis (2023) are solved by our real-time schedule slot checker, which mirrors the live tracking and queue-control methodologies studied by Salokhe et al. (2026) to manage the kitchen's daily capacity and stop timeline errors.

The second major area discussed by the researchers is market competitiveness and business insight. To ensure a pastry business remains competitive, the local study by Omowole and Sari (2025) emphasizes that modern Filipino consumers demand clear pricing layouts and active digital engagement. Our project directly applies this insight by introducing an interactive Custom Cake Builder with real-time price estimation alongside clear menu categories. This removes purchasing friction for the customer. Omowole and Sari (2025) also note that independent bakeries struggle because they lack data collection software to

monitor their sales trends. This local struggle matches the findings of the foreign study by Martinez (2024), which points out that bakeries globally lose significant profit margins because they do not track their best-selling items or financial metrics digitally.

We solve this exact problem by applying the data aggregation and sales summary principles discussed in the related literature by Azzahra et al. (2025). Our system automatically processes raw transaction inputs in the background, transforming cold data rows into visual analytics complete with monthly sales charts and top-product graphs, giving the owner instant business insights without manual arithmetic errors.

Finally, the technical execution of our platform relies heavily on the cloud frameworks and real-time networking principles established in the reviewed literature. The technical research by Tennakoon et al. (2022) and Salokhe et al. (2026) supports our choice of a cloud-native, serverless NoSQL database model using the free Firebase Spark Plan. This ensures that C&N Bakery operates with zero monthly hosting costs while maintaining lightweight JSON text structures that scale smoothly during holiday traffic surges. High-volume media files, such as payment receipts and design reference photos, are offloaded to Facebook Messenger to protect database limits.

Once orders are safely saved, the system keeps the entire business connected by using the live event-driven synchronization models detailed in the dashboard applications from Salokhe et al. (2026) and Azzahra et al. (2025). The moment a customer submits an order online, a live update syncs with the back-office dashboard. Kitchen staff can then seamlessly track and move the order through our 4-stage stepper .

In summary, by piecing all foreign studies, local studies, and related literature together, it is clear that transitioning from a manual, chat-based workflow to an integrated, real-time web application is necessary for C&N Bakery. Our Capstone project bridges the gap between traditional baking and modern software engineering, creating an automated ecosystem that protects profit margins, eliminates communication errors, and streamlines the bakery's daily workflow.

CHAPTER 3

3.1 Research Design

For this project, we chose a descriptive-developmental research design. This approach combines two straightforward steps: first, studying a real-world problem in detail to understand exactly how it works (descriptive), and second, building a practical software tool to fix it (developmental).

By combining these two methods, we made sure we didn't just guess what features to code. Instead, every feature on our website is designed to solve a real, everyday struggle that the bakery faces, built using reliable coding practices.

1. Descriptive Research

Descriptive research is all about looking closely at a real situation exactly as it is right now, without trying to change anything. Instead of testing abstract theories in a lab, this method takes a clear, accurate snapshot of daily operations. It maps out how tasks are done, documents the work habits of the staff, and points out the exact bottlenecks that slow things down. It answers the basic questions of what, who, where, and how a process works so that we don't make wrong assumptions when writing code.

- How We Used It in Our Project:

We used descriptive research to study, understand, and document the manual struggles of C&N Bakery before writing a single line of code. We spent time at the bakery observing daily counter work, interviewing the business owner, and looking over their physical paperwork. This step revealed four major pain points:

- *Messy Order Tracking:*

Customer inquiries and order requests were scattered everywhere—across Facebook Messenger chats, SMS texts, and phone calls. Staff had to scroll through endless, unorganized chat histories just to find a single design idea, price quote, or delivery date.

- *Lost Order Details:*

Complex cake details like flavor choices, icing types, sizes, and custom name inscriptions were easily miscopied onto loose scraps of paper or forgotten completely during busy kitchen shifts, leading to costly mistakes.

- *Calendar and Delivery Confusion:*

Pickup dates and delivery schedules relied entirely on a handwritten notebook. Without an automated calendar to protect them, the bakery suffered from human scheduling mistakes, accidental double-bookings, and immense stress during major holidays.

- *Exhausting Financial Reviews:*

Tracking down-payments, verifying GCash transfers, and adding up cash numbers were done manually with calculators and paper ledgers. This manual math routine took up too much time and increased the risk of accounting errors.

- *Why It Matters to Our Study:*

Without this descriptive phase, our web platform would have no real-world grounding. This step allowed us to turn general business complaints into exact software features. It helped us map out exactly how money changes hands and calculate the physical workload the kitchen staff can handle, ensuring our final application solves real business problems.

2. Developmental Research

Developmental research is the step-by-step process of designing, building, coding, testing, and perfecting a software application over a set timeline. The main goal of this method is to take the raw notes and problems gathered from the descriptive phase and turn them into a practical, working tool. It is a continuous loop where early versions of the website are built, tested, reviewed by the actual users, and adjusted until the software runs smoothly and reliably under real-world conditions.

- How We Used It in Our Project:

Developmental research is the technical core of this project. This is the phase where we shifted away from just talking about problems and actually started writing code to build a real solution. Using the exact list of complaints and rules we gathered from our descriptive observations, our team designed and programmed a web-based platform featuring two main sections connected to a single cloud database:

- *The Customer Portal:*

A public online storefront with an easy-to-read menu split into clear categories. To help budget-conscious

buyers, we added simple price range filters. We also coded a live price calculator, which automatically estimates the baseline price as customers mix and match cake tiers, sizes, and flavors. Finally, we added a secure tool for customers to upload reference photos, a clear text field for cake inscriptions, and a self-service page to track their order status.

- *The Management Dashboard:*

A secure administrative workspace protected by a login screen. Inside, we built dashboard summary cards that give the owner a quick look at daily business activities. We also added visual data tools (like graphs and charts) to summarize monthly sales and track which products are selling best.

- *System Automation Logic:*

We programmed background rules to make both portals talk to each other safely. The most important rule is our smart booking calendar, which counts incoming orders and automatically blocks out public booking dates once they hit a maximum limit set by the owner. We also integrated interactive progress controls so kitchen staff can easily update an order's status as

it moves through preparation, baking, and completion milestones.

- Why It Matters to Our Study:

The developmental phase is what gives our project its value in the field of information technology. It serves as a real-world example of how affordable cloud database networks can be customized to support small, local neighborhood bakeries. By documenting how smart coding rules can stop everyday human mistakes—like shutting down a calendar date when the kitchen is full or putting order details right next to customer reference images—this phase proves how custom-made software can protect and upgrade small retail businesses.

3. Why We Did Not Use Experimental Research

Experimental research is a strict lab method used to find cause-and-effect links. It works by changing one specific detail (a variable) while keeping everything else exactly the same, then measuring the result using complex statistics and control groups.

- Why It Does Not Fit Our Project:

This method does not work for our study for a few simple reasons:

- *No Variables to Change:*

We are not running laboratory tests to see how changing oven heat levels or ingredient brands affects baking speeds. Our actual goal is to build a working web application that fixes real office problems.

- *Building Over Theory:*

This is an IT project focused on building a useful tool, not a science experiment. Success means making sure the website is safe, fast, and easy for C&N Bakery to use, not testing abstract math formulas.

- *Bad for Daily Business:*

Splitting real bakery customers into an "unlucky control group" forced to use slow, messy paper notes and a "lucky test group" using the new website just to measure their frustration would disrupt daily shop sales and annoy buyers.

- *Focus on System Integration:*

Since the project is meant to replace an old, manual tracking style with an all-in-one cloud setup, the correct path must be developmental. We need to focus on coding secure database gates and price calculators, not running clinical lab controls.

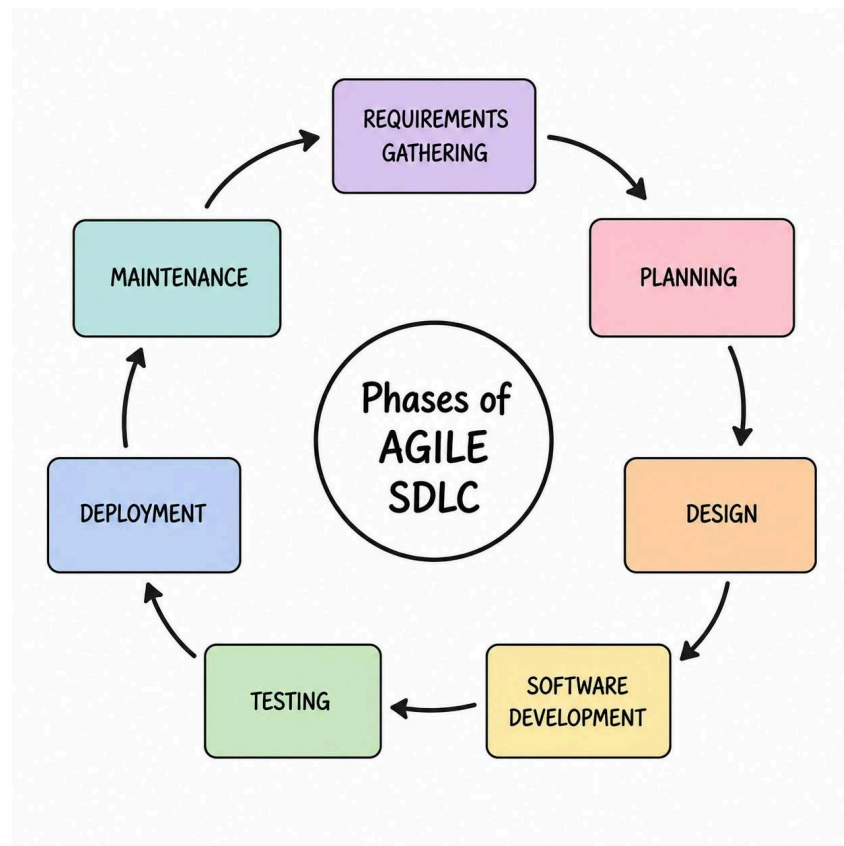
3.2 System Development Methodology

For this project, the research team selected the Agile Model as the primary framework for the Software Development Life Cycle (SDLC). Traditional software methods, like the classic Waterfall model, follow a rigid, linear path where each phase must be completely finished before the next one can begin. In a Waterfall approach, returning to modify a previous phase is highly difficult and expensive once coding has started.

However, building software for local micro-enterprises like C&N Bakery requires a much more flexible approach. As the bakery owner, kitchen staff, and everyday customers interact with early prototypes of the web application, they will inevitably discover new physical operational realities and workflow needs that were not apparent at the beginning of the project.

The Agile methodology solves this problem by focusing on flexibility, constant communication, and delivering working pieces of the system in short, focused blocks of time called "sprints." Instead of waiting until the final defense to see the finished website, the bakery management can inspect, test, and provide real-world feedback on small, working parts of the application as they are being built. This continuous feedback loop minimizes development risks, prevents the team from wasting

time building unnecessary features, and ensures that the final software smoothly integrates into C&N Bakery's day-to-day kitchen operations.



1. Requirements Gathering

- **Definition:**

This phase is all about figuring out exactly how the website needs to work and perform. The team writes down a clear checklist of every button, page, security lock, and performance standard needed to solve the business's

problems. This process translates basic human business needs into logical guidelines that programmers can follow. Without a detailed requirements checklist, developers risk spending weeks building a website that runs perfectly but does not actually solve the customer's real-world issues.

- **How We Plan to Apply It:**

The research team will conduct structured consultation sessions with the owner of C&N Bakery to decide exactly what the website must do and what it must not do. The proposed system will focus directly on fixing the bakery's biggest operational struggles: letting customers design cakes online, calculating custom prices automatically, protecting the kitchen calendar from overbooking, and keeping track of daily sales.

To keep the project realistic and deliver it on time, complex, expensive features that the bakery does not need right now—like automated delivery truck GPS routing or automatic wholesale ingredient ordering—will be explicitly excluded. Through these planning sessions, the team will establish two master checklists:

- What the system must do:

The customer-facing website will feature an organized menu divided into clear categories so users can browse items easily. It will include a step-by-step cake design tool that calculates an estimated price instantly as the user chooses their cake size, layers, and frosting type. It also requires a secure upload utility for design photos, a text box where users can type custom cake lettering, and a personal tracking page where customers can check their order status. For the admin side, the system will provide a master order list with search and filter bars, simple progress buttons for the kitchen staff, and visual charts showing daily sales trends.

- How the system must perform:

The website will utilize a responsive design, meaning it must adjust its layout automatically to look clean and work smoothly on any smartphone, tablet, or desktop computer. For security, the admin dashboard will be locked behind a secure login screen to keep business records private. For performance, the database must load and save information quickly, and the pages must load fast even on slower mobile data connections.

2. Design

- Definition:

The design phase is where we turn our feature checklist into visual and logical blueprints before writing any actual code. This involves sketching user interface (UI) wireframes to plan where things go on the screen, and structuring database layouts to show how different pieces of information will connect, talk to each other, and save securely in the backend. In Agile development, this phase is like drawing a house plan before buying the wood. It helps developers catch logical mistakes on paper first, saving hours of coding time and preventing messy re-writes later.

- How We Plan to Apply It:

The research team will map out the complete plan for the C&N Bakery platform before coding begins. We will draw clean wireframes to plan where navigation bars, cake selectors, shopping carts, tracking steps, and sales graphs will sit on the screen. At the same time, we will structure our database schema to keep all data organized and avoid duplicate records:

- Linking Order Details:

The team will design the database so that a customer's custom cake choices, written notes, and uploaded design photos all bind securely to one single, unique Order ID. This is planned to prevent details from getting mixed up or lost in the system.

- Calendar Capacity Setup:

We will structure the database rules to actively count how many orders are placed for any given date. This is designed to give the system the logic it needs to recognize when a calendar date has hit its maximum limit of 5 orders, allowing it to block future checkouts for that day.

- Sales Data Organization:

The team will arrange the transaction files using clear status labels . This structured layout is planned to ensure the admin dashboard can easily pull and display clean sales metrics without causing system lag.

3. Implementation

- Definition:

This is the phase where the actual building and coding happen. Programmers write the functional code, build the visual screens, connect the pages to the database, and turn the blueprints into a real website that users can click on and use. In the Agile model, development is not done all at once in one long, exhausting coding marathon. Instead, it is broken down into separate sprints where small, manageable features are coded, reviewed, and integrated step-by-step. This helps developers spot and fix coding conflicts early before they turn into bigger system bugs.

- How We Plan to Apply It:

The team plans to write the web application using standard, reliable front-end web tools (HTML5, CSS3, and JavaScript) and connect them to a secure cloud database for real-time data storage. Following Agile practices, we will divide our coding schedule into three distinct sprints:

- Front-End Customer Portal :

This will focus on coding the public website. The team plans to program the responsive menu, the step-by-step cake customizer , the photo upload utility for custom designs, and the personal order tracking page.

- o Back-End Management Dashboard :

This sprint will focus on coding the private admin workspace. We plan to program the login security gate, the capacity logic that checks the database and automatically blocks checkout dates once they hit the limit of 5 orders, and the simple progress buttons that let kitchen workers change order statuses with one click.

- o System Integration :

This final development block will focus on connecting the frontend and backend. The team plans to write queries to calculate raw sales numbers into clean, visual graphs for the owner, and set up the automated system rules so that any status update made by the admin instantly updates the customer's tracking page.

4. Testing

- Definition:

This phase defines the step-by-step plan we will use to check that the website works correctly, is secure, and is stable under different conditions. This phase outlines the exact testing methods and passing criteria we established to find bugs and verify the system before letting the bakery use it. Writing this plan beforehand ensures our testing is organized, fair, and focuses on making sure every feature works exactly the way it is supposed to.

- How We Plan to Apply It:

To ensure the platform is completely reliable, the team will structure a multi-layered testing plan to check the system's primary features:

- Unit Testing Plan:

The team will establish steps to test individual pieces of code on their own. This will include entering different combinations into the cake builder to make sure the pricing script calculates costs perfectly every time, entering invalid phone numbers

to check if the form blocks them, and verifying that the login screen blocks incorrect admin passwords.

- Integration Testing Plan:

We plan to establish steps to verify that separate parts of the website work together. This will include checking that when a user uploads a design photo, the file saves safely in cloud storage and its web link maps correctly to the user's invoice in the database, and ensuring that changing an order status on the admin panel instantly updates the customer's screen.

- End-to-End System Testing Plan:

The team will design testing scenarios to simulate complete customer journeys from start to finish—from logging in, customizing a cake, uploading a payment receipt, and checking out, to checking if the order shows up on the admin ledger. This plan also includes setting a test date to its maximum capacity of 5 orders to verify that the system successfully blocks the 6th customer from selecting that date.

- User Acceptance Testing (UAT) Plan:

We will structure a feedback guide for C&N Bakery's actual stakeholders (the owner, the baking staff, and a group of everyday customers). This protocol details how the team will measure how easy the website is to use, how fast users can complete tasks, and whether the staff can manage orders without confusion in a controlled test environment.

5. Deployment

- **Definition:**

This phase outlines the steps for putting the tested website live on the internet, setting up the cloud databases, and officially preparing for the launch. A clear deployment plan is vital for small businesses because it outlines how to launch the new system smoothly without interrupting their daily operations.

- **How We Plan to Apply It:**

Our deployment plan outlines the exact sequence of steps needed to launch the website and prepare the system for the bakery's live use:

- Database Cleanup and Setup:

Before going live to the public, the database is planned to undergo a complete cleanup to delete all mock test accounts, fake custom orders, and test photos used during our development and testing phases. The team will then load the fresh database with C&N Bakery's actual menu items, verified prices, and real product categories.

6. Maintenance

- Definition:

In the Agile model, deployment is not the final step of the lifecycle. The maintenance phase represents the post-launch support, monitoring, and evaluation stage. Once the software is in the hands of real users, the team collects operational feedback from the client and customers to identify new bottleneck issues, record bug reports, and plan minor adjustments for the next development loop. This continuous maintenance cycle ensures the software can evolve alongside the business.

- How We Plan to Apply It:

A post-launch feedback and maintenance protocol is planned to manage the live platform:

- Performance Monitoring:

Plans have been outlined to monitor database response times and verify that the calendar booking logic actively blocks overbooking under real live traffic.

- Owner and Staff Feedback Loop:

A structured feedback loop will be created where the bakery owner and kitchen staff can note any operational issues or feature suggestions during their first few weeks of live use. This feedback will be documented to guide any future software updates or maintenance cycles, ensuring the platform remains highly practical and secure.

3.3 System Architecture

The proposed system will utilize a Serverless Cloud Architecture powered by the Google Firebase Spark (Free Tier) Plan. Unlike traditional client-server models that require a custom backend stack (such as a local server running PHP and a separate database engine), this architecture allows the front-end interface to communicate directly with cloud services through secure, built-in network channels.

By using a serverless model on Firebase's free tier, the system eliminates administrative hassle, hardware maintenance costs, and server hosting fees. To stay strictly within Firebase's 1 GB free database storage ceiling, the system architecture is intentionally designed to offload high-volume media storage entirely. Instead of storing large image files (such as payment receipts or design references) on expensive cloud servers, customers submit visual proofs directly through the bakery's official Facebook Page Inbox (Messenger). This keeps the cloud database lightweight, storing only structured text records (like user profiles, cake parameters, order schedules, status logs, and sales metrics).

The architecture is divided into two main environments:

1. The Client-Side Application

- **Definition:** The client-side represents the visual front-end of the web application—the screens, buttons, forms, and interactive menus that users see and interact with inside their web browsers on mobile phones, tablets, or computers.
- **Application:** The front-end for both the Customer Portal and the Admin Dashboard is built using HTML5 for structure, CSS3 for responsive styling, and Vanilla JavaScript for interactive logic. The client-side executes core interactive tools—such as the Custom Cake Builder, menu filtering, and real-time price estimation—directly on the user's device. Running the price calculation engine locally in the browser gives customers instant, lag-free cost updates as they customize their cake tiers, flavors, and decorative elements.

2. The Backend-as-a-Service (BaaS) Layer

- **Definition:** A Backend-as-a-Service (BaaS) model allows the application to connect directly to ready-made cloud infrastructure instead of building and maintaining a custom backend server from scratch. The cloud provider manages background tasks like user authentication, data synchronization, and cloud security, enabling fast and cost-effective development.

- **Application:** Google Firebase serves as the core cloud engine, handling security and data storage through two essential free-tier modules:

A. Firebase Authentication

- **Definition:** A cloud-hosted identity service that manages user sign-ups, login verification, password encryption, and account access permissions.
- **Application:** This service acts as the platform's security gatekeeper. It manages customer account creation, login sessions, and administrative clearance. When a user attempts to access the Admin Dashboard, view financial analytics, or update store settings, Firebase Authentication verifies their admin credentials. If a standard customer tries to access restricted administrative routes, the system automatically blocks them, keeping store records and financial reports secure.

B. Firebase Realtime Database

- **Definition:** A cloud-hosted NoSQL database engine that organizes application data as a JSON tree and syncs updates live across all connected devices without requiring manual page refreshes.

- **Application:** This module acts as the central data store for the entire platform. It maintains lightweight JSON trees for user profiles, product catalogs, custom cake options, active orders, fulfillment schedules, and aggregate sales records. Because it uses real-time synchronization, the moment an admin approves an order or updates a status on the 4-stage stepper the changes reflect live on the customer's tracking dashboard. It also enforces daily schedule limits by dynamically counting orders placed for a specific date and locking out booked-out slots to prevent overbooking.

3.4 Tools and Technologies Used

To construct, deploy, test, and manage the serverless platform, a specific stack of programming technologies, development tools, and database frameworks was chosen. The following sections detail each tool, outlining its theoretical definition and how the research team plans to apply it to build the C&N Bakery platform.

1. HyperText Markup Language (HTML5)

- **Definition:**

HTML5 is the standard foundational language used to build and arrange content on the web. It uses semantic tags to tell the web browser exactly where to place elements like text headings, entry boxes, buttons, media blocks, and data grids.

- **How the Team Will Apply It:**

HTML5 will act as the structural skeleton of the entire application. It is planned to establish the input forms inside the custom cake builder, set up the semantic document frames for the order viewer, and create the structured data tables for the administrator control panel. This ensures that raw database strings—like specific cake

text inscriptions, customer phone numbers, and transactional histories—will be presented clearly and accurately to both the bakery staff and public customers.

2. Cascading Style Sheets (CSS3)

- Definition:

CSS3 is a design styling language used to control the overall look of a website. It manages color schemes, typography, margins, layout padding, and how elements shift automatically to fit different device screens.

- How the Team Will Apply It:

CSS3 will control the visual presentation and responsive behavior of the public storefront and private management dashboard. It will handle the mobile-friendly design, automatically turning a dense administrative desktop layout into a simplified, cleanly stacked vertical interface on smaller mobile devices. It will also style the interactive kitchen tracking timeline, make live pricing adjustments stand out visually, and apply the color-coded labels that instantly separate paid receipts from pending cash collections.

3. Vanilla JavaScript

- Definition:

JavaScript is the core browser programming language used to build interactive features, adjust web document structures on the fly, run client-side mathematical calculations, and exchange data over the network without relying on bulky external software packages.

- How the Team Will Apply It:

JavaScript will serve as the operational engine running the front-end of the website. It will drive the live calculator script, instantly changing the checkout quote as a customer updates different design fields like sizes and flavors. It will also scan order records to count active submissions per date to power the calendar locking gates, and run the quick text-search filters on the management tables. Most importantly, it will handle the direct integration with our cloud backend, converting text inputs and image files into formatted data streams without needing a traditional server-side language like PHP.

4. Firebase Authentication

- Definition:

A cloud-hosted identity validation service that provides built-in encryption, user session management, token handshakes, and access rule checks to secure personal profiles.

- How the Tool Will Be Applied:

Within the proposed development stack, this tool will be utilized to deploy a pre-configured security layer without the need to code custom password hashing algorithms. The team will use its SDK integration to build the system's user verification gates, ensuring that when an individual attempts to access private sales charts, view financial records, or toggle a menu item's availability, the application will be able to securely check their account rank and shield backend metrics from unauthorized access.

5. Firebase Realtime Database

- Simple Definition:

A cloud-hosted, highly scalable NoSQL database engine that structures application information as an organized JSON

tree and synchronizes state updates instantly across all connected screens using live web socket communication lines.

- How the Tool Will Be Applied:

The research team will select this framework to handle the application's central data storage needs. In the development environment, it will allow the team to model data flexibly without managing complex SQL tables or writing long database queries. It will store menu options, product categories, buyer history files, custom cake configurations, and scheduling rules as a structured data tree. Because it syncs in real time, it is designed to allow new orders to instantly appear on the management dashboard the moment a customer checks out, without requiring a manual screen refresh.

6. Firebase Backend Infrastructure (No-Cost Spark Plan)

- Definition:

The no-cost tier of Google Firebase's Backend-as-a-Service (BaaS) platform, providing cloud hosting, user authentication, and real-time database capabilities within defined free usage limits.

- How the Team Will Apply It:

The research team will use the Spark Plan to host the web application and run the backend cloud services completely free of charge. To ensure the platform operates smoothly without exceeding the 1 GB database storage ceiling, the system architecture is designed to store only lightweight, structured text data—such as user accounts, cake options, order details, schedule counts, and sales metrics.

High-volume media files, including cake design reference photos, keeping the Firebase backend fast, lightweight, and 100% cost-free.

7. Visual Studio Code (VS Code)

- Definition:

A powerful, lightweight source-code editor and Integrated Development Environment (IDE) featuring smart syntax highlighting, automatic text formatting, built-in error tracking, and integrated workspace controls.

- How the Team Will Apply It:

VS Code will serve as the primary engineering workspace used to write and organize the platform's source files. The team will use its programming tools to draft HTML layouts,

style CSS rules, and program JavaScript logic. Its built-in code checking is planned to warn developers about missing variables or punctuation typos immediately, keeping the software code clean and easy to maintain throughout each development sprint.

8. Git and GitHub

- Definition:

A distributed version control tracking ecosystem paired with a cloud-hosted code repository service, used to log individual software updates, preserve safe history snapshots, and manage collaborative team coding projects.

- How the Team Will Apply It:

Git will be used to log every development update made to our project's codebase, serving as a reliable backup network. Every time a major system module is completed, a safe snapshot of the working code will be sent to GitHub. This will allow the team to test out new code modifications safely without risking previously working features, keeping a clean, historical timeline of our software development as we work through our sprint schedule.

9. Firebase Console Dashboard

- Definition:

A centralized, web-based management control panel provided by Google that lets developers inspect database traffic, review user accounts, configure security boundaries, and monitor live system performance logs.

- How the Team Will Apply It:

The development team will use this dashboard to oversee our underlying cloud networks. It will let us write database data access rules, monitor cloud storage space usage, and review real-time server activity logs. If an execution issue or file upload error occurs during development testing, the console will display detailed logs that point out exactly where the code or network rule failed so the team can address it immediately.

3.5 Data Gathering Techniques

1. Interviews

- What It Means:

An interview is a direct conversation where a researcher asks structured or open-ended questions to individuals or groups to gather detailed personal insights, firsthand experiences, and specific user requirements.

- How We Plan to Apply It:

The development team will conduct structured consultation sessions with the C&N Bakery owner, front-counter staff, and kitchen bakers. These conversations will focus on understanding how order costs are calculated, the operational frustrations caused by unconfirmed digital payments, and the maximum volume of complex custom orders the kitchen can safely execute in a single day. These technical requirements will serve as the foundational logic for the system's core capabilities.

- Planned System Integration:

- Scheduling Rules:

The owner's input regarding physical daily production limits will be translated into system rules, where a

backend data counter tracks accepted orders per date and blocks new checkout requests once the daily limit is reached.

- Payment Verification Rules:

To address counter staff struggles with manual digital bank transfers, the team plans to establish a requirement for digital payment proof. The checkout process will be designed to require customers to upload their digital payment receipts as part of their order details.

- Order Customization Rules:

To prevent last-minute order modifications from being lost in personal text messages, the system will be planned to collect all custom requests and text inscriptions directly at the point of ordering, saving them together under a single order record.

2. Observation

- What It Means:

Observation is a data gathering method where researchers watch real-world activities as they unfold in their natural

setting to record actual behaviors, operational timelines, and procedural habits without interfering with the workflow.

- How We Plan to Apply It:

The team plans to spend time inside the active bakery environment during peak operational hours to shadow workers as they take manual counter orders, track active baking tasks within the kitchen, and calculate daily sales totals. By watching these workflows firsthand, the team aims to identify critical operational bottlenecks, such as staff losing track of an order's physical progress or writing raw tallies down on loose paper pads.

- Planned System Integration:

- Order Lifecycle Monitoring:

Observing how the kitchen staff tracks task completion is planned to inform the layout of the management workflow. The system will be structured to track orders through sequential progress states so staff can see which cakes are pending, in-progress, or ready for pickup.

- Sales Tracking Framework:

The team will analyze how the owner compiles paper bills to calculate daily profits. This workflow will guide the requirements for the system's reporting capabilities, ensuring the database organizes transactions in a way that allows daily sales summaries to be generated automatically.

- Item Filtering Framework:

To streamline the ordering process for budget-conscious walk-in customers, the storefront is planned to support price-range grouping, allowing users to view items that fit their specific spending capacity.

3. Document Analysis

- What It Means:

Document analysis is the systematic review and evaluation of an organization's existing physical records, historical logs, and business forms to understand their baseline rules, information structures, and standard administrative formatting.

- How We Plan to Apply It:

The development team will gather and cross-reference C&N Bakery's physical paper logbooks, handwritten customer order slips, official base pricing reference sheets, and product catalogs. Reviewing these physical artifacts is planned to provide the exact mathematical steps, structural layouts, and business rules that the bakery has successfully relied on for years.

- Planned System Integration:

- Ordering:

The structural layout of the bakery's handwritten paper order forms (such as size, flavor, and icing fields) will serve as the blueprint for designing the digital order input parameters.

- Pricing Logic:

The team will analyze the bakery's handwritten pricing sheets to map out the mathematical formulas needed for the system's calculation logic. This will ensure that combinations of sizes, flavors, and design add-ons generate quotes that match the bakery's official pricing structure.

- Database Organization Scheme:

Historical product catalogs and logs will be used to design the database architecture, ensuring that items are organized into distinct, logical categories.

- Order Status Categorization:

Reviewing older paper receipts revealed that the bakery struggled to distinguish between fully settled transactions, partial down-payments, and pending cash collections. To resolve this, the system's database schema is planned to categorize transactions using standardized status indicators, replacing the staff's historical habit of marking paper records with colored highlighters.

3.6 Testing Procedures

To ensure that the C&N Bakery website operates correctly, keeps transaction data secure, and is fully prepared for daily business operations, the research team will execute a structured, multi-phase testing methodology. The testing process will evaluate everything from the client-side mathematical scripts that update cake prices on the screen to the real-time database synchronization channels that link the storefront with the admin dashboard.

The following sections detail each planned testing phase, its procedural definition, and the specific application protocol developed for the proposed system:

1. Unit Testing

- **What It Means:**

Unit testing is when you isolate and check the smallest individual pieces of a website's code—like a single button, text input box, or mathematical formula—to make sure that specific piece works perfectly on its own before connecting it to the rest of the system.

- Planned Testing :

The team will isolate and independently evaluate the system's modular micro-functions:

- Cost Calculator Verification: The mathematical calculation script running inside the cake builder will be tested by simulating different combinations of cake sizes, layers, and frosting types. This is designed to verify that the formula consistently computes the correct pricing outputs.
- Budget Filter Execution: The filtering logic will be tested by triggering each predefined price bracket to ensure that the system accurately isolates and displays only the items that fall within that specific budget range.
- Authentication Security Verification: The login form inputs will be evaluated using boundary value testing—such as submitting empty fields, entering incorrectly formatted email structures, and entering invalid password lengths. This will ensure that the system successfully blocks unauthorized access attempts and displays appropriate error warnings.

2. Integration Testing

- **What It Means:** Integration testing is when you combine different parts of the website that were built separately and test them together as a group. This makes sure they can talk to each other, share information smoothly, and work as a team without causing glitches.
- **Planned Testing Protocol:** The team will execute integration tests to verify that the communication channels between the client-side interface and the Google Firebase cloud services function seamlessly:
 - **File and Record Link Testing:** This test will verify that when a customer uploads an image file (such as a payment receipt or custom cake sketch), the storage service successfully saves the file, generates a secure URL, and writes that link directly into the corresponding text-based order row in the database.
 - **Automated Booking Controls:** The team will test the connection between the checkout interface and the database calendar logs. This is planned to ensure that the exact second an order is completed, the background data counter updates the daily limits in real time.
 - **Live Status Syncing:** We will test the persistent web socket connection between the admin panel and the

customer's tracking view to ensure that order status changes made by the staff are reflected on the customer's interface immediately without requiring a manual browser refresh.

3. System Testing

- **What It Means:** System testing is when you test the entire website from beginning to end as a single, complete package. This ensures that the whole journey—from a customer placing an order to the shop owner receiving it and tracking the business numbers—works flawlessly without any broken links or crashes.
- **Planned Testing Protocol:** The team will evaluate the platform as a fully unified system to verify the complete operational workflow:
 - **End-to-End Business Cycle Simulation:** The team will execute complete ordering simulations, starting from user authentication, custom cake building, payment proof uploading, and calendar date selection, and ending with verifying that the entire record registers correctly on the management panel.
 - **Capacity Safeguard Verification:** The scheduling calendar's structural rules will be tested by repeatedly submitting mock orders on a single date

until the daily cap is reached. This is designed to verify that the system automatically locks the date, making it unselectable for subsequent transactions.

- Data and Workflow Flow: The team will move test records through every sequential status step to ensure that when an order is flagged as completed, the reporting database automatically updates the corresponding sales metrics and dashboard summaries.

4. User Acceptance Testing (UAT)

- What It Means: User acceptance testing is the final evaluation phase where the actual target audience—including business owners, managers, staff, and end-consumers—interacts with the system to assess its usability, operational efficiency, and overall value.
- Planned Testing Protocol: The research team will establish a controlled environment to gather direct feedback from C&N Bakery's key stakeholders:
 - Customer Usability Sessions: A small testing group of regular customers will be asked to complete baseline tasks, such as navigating the menu, designing a cake, and checking out, while the research team observes their interactions to evaluate navigation clarity and ease of use.

- Administrative Workflow Integration: The bakery owner, counter staff, and kitchen bakers will interact with the administrative management interface. This test is designed to evaluate how easily staff can review incoming custom orders, verify uploaded payment documents, toggle inventory items, and advance tasks through the production pipeline, ensuring the software fits smoothly into their real-world routines.

3.7 Evaluation Criteria

To determine the overall success, stability, and operational value of the proposed C&N Bakery platform, the system is planned to be assessed against a definitive set of performance criteria. These metrics will provide a structured framework to evaluate the technical health of the cloud architecture and the actual user experience once development is complete.

Each evaluation criterion is detailed below, highlighting its fundamental meaning, how it applies to the architecture, and the concrete operational details to be verified during the evaluation phase:

1. Functionality

- **What It Means:** Functionality checks whether a software application completely and accurately performs every single feature, calculation, and administrative task it was intentionally designed to execute.
- **Planned Application:** The research team will evaluate the system to confirm that both the public storefront interface and the private management control panel execute their respective digital tasks exactly as specified in the project requirements.

- **Operational:** On the customer side, the evaluation will verify that the catalog menu successfully applies budget-based filtering categories and that the cost calculator computes live pricing updates accurately as different custom specifications are selected. The team will confirm that automated system status notifications trigger successfully upon order placement and production milestone changes. On the administrative side, we will verify that the analytical dashboard generates real-time revenue and product popularity charts. Furthermore, the evaluation will check that the interactive management panel seamlessly advances order records through the multi-stage production tracking lifecycle without layout errors.

2. Usability

- **What It Means:** Usability measures how easy, intuitive, and stress-free a website is for real people to navigate, understand, and interact with on their own without requiring specialized technical training.
- **Planned Application:** The evaluation will measure how quickly and comfortably everyday consumers and busy bakery employees can complete their essential tasks on the platform using different types of devices.

- **Operational:** For customers, usability testing will ensure they can seamlessly select a product, type out custom design text instructions, pick an open calendar date, and attach a digital payment confirmation file all on a unified checkout flow. For the bakery personnel, the team will evaluate the layout's readability under active kitchen conditions. The evaluation is designed to check if the responsive framework automatically transforms dense data grids into clean, highly legible lists when loaded onto portable display terminals, allowing staff to tap tracking controls or toggle ingredient stock availability indicators with a single touch.

3. Reliability

- **What It Means:** Reliability evaluates how consistently a software system runs over extended periods without crashing, dropping critical information, or freezing, as well as how smoothly it recovers from temporary network interruptions.
- **Planned Application:** The research team will verify that the system's cloud connection remains completely stable and that transaction data is safely preserved even during sudden internet drops or moments of high user traffic.

- **Operational:** The team will test the system by submitting multiple concurrent order transactions to ensure the serverless backend handles simultaneous requests smoothly without performance degradation. Because the architecture is planned to operate on managed cloud infrastructure, the evaluation will verify that it scales dynamically to process data spikes without slowing down. We will also confirm that if a user experiences an abrupt disconnection mid-checkout, their profile attributes and order details are preserved safely in the cloud repository. The evaluation will observe whether the real-time data sync re-establishes itself automatically when connection lines drop momentarily, keeping the manager's live display screen active.

4. Efficiency

- **What It Means:** Efficiency measures the speed, responsiveness, and performance of a system, focusing on how quickly it processes database updates, handles text search queries, and automates manual workflows.
- **Planned Application:** The research team will measure how much faster the automated cloud system performs daily shop operations compared to the bakery's historical reliance on manual paper logbooks and physical planners.

- Operational : The team will monitor data transmission latency to confirm that the exact millisecond a customer finalizes a transaction at home, the new order entry appears on the kitchen's management board without requiring a manual browser page refresh. We will also verify that the administrative search tools filter through hundreds of transaction rows instantaneously as the user types. Additionally, the evaluation will check if the database's capacity tracking logic calculates daily bookings in real time, locking out fully booked dates instantly.

5. Security

- What It Means: Security encompasses the defensive structures, cloud access controls, and input validation rules used to safeguard sensitive personal information, user credentials, and financial files from unauthorized access or digital exposure.
- Planned Application: The team will evaluate the system's defensive configurations to guarantee that customer profiles, internal business metrics, and financial confirmation records stay fully protected.
- Operational Benchmarks: The evaluation will verify that the authentication layer successfully acts as a digital gatekeeper, validating account clearance on the cloud side

so that standard customer accounts are completely blocked from accessing or modifying administrative database paths. The team will also test the system's text validation parameters to ensure that unauthorized code scripts cannot be executed within public input forms or text boxes. Finally, we will confirm that uploaded digital payment documents are archived within a secure cloud storage bucket accessible only via verified administrator sessions, ensuring financial data privacy.

CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 Overview of the Developed System

The C&N Bakery Custom Cake Ordering and Management System is a mobile-responsive web app designed to make custom cake ordering smooth for customers and easy to manage for bakery staff. The system brings order customization, booking schedules, payment verification, and order tracking together into one digital platform.

Unlike basic online menus or generic shop templates, the platform offers an interactive Custom Cake Builder. Customers can select cake sizes, tiers, flavors, and design features while seeing instant price updates. To prevent overbooking, the system checks daily production limits in real time as orders are placed. To keep cloud costs low on the free Firebase database tier, payment proofs (like GCash or bank receipts) are sent directly to the official C&N Bakery Facebook page for manual staff verification.

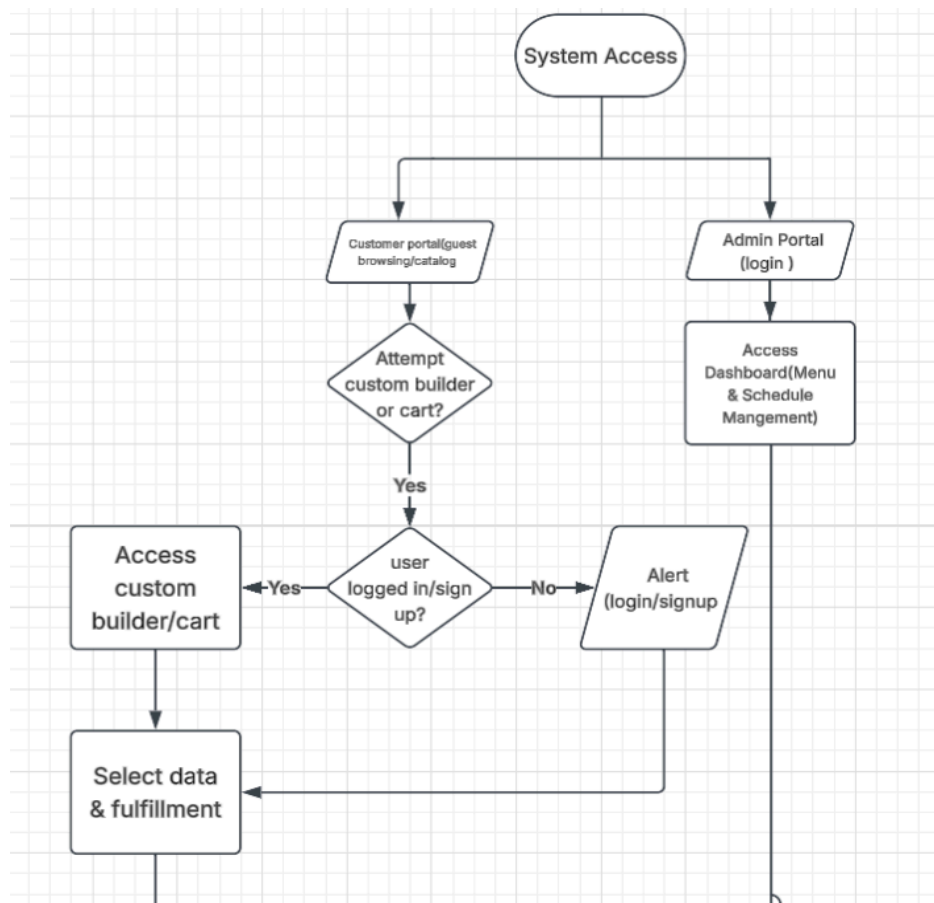
The primary goal of the system is to help C&N Bakery reduce miscommunications, save staff time, and give customers clear, transparent updates from order to delivery.

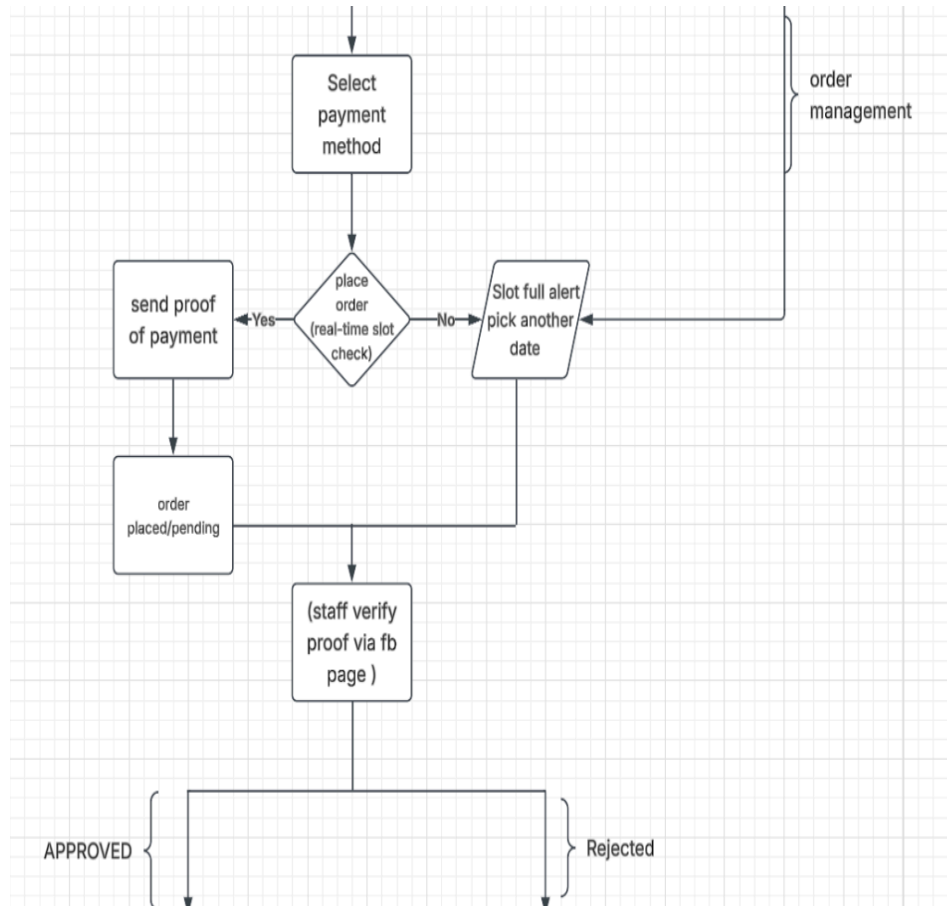
The major objectives of the developed system are as follows:

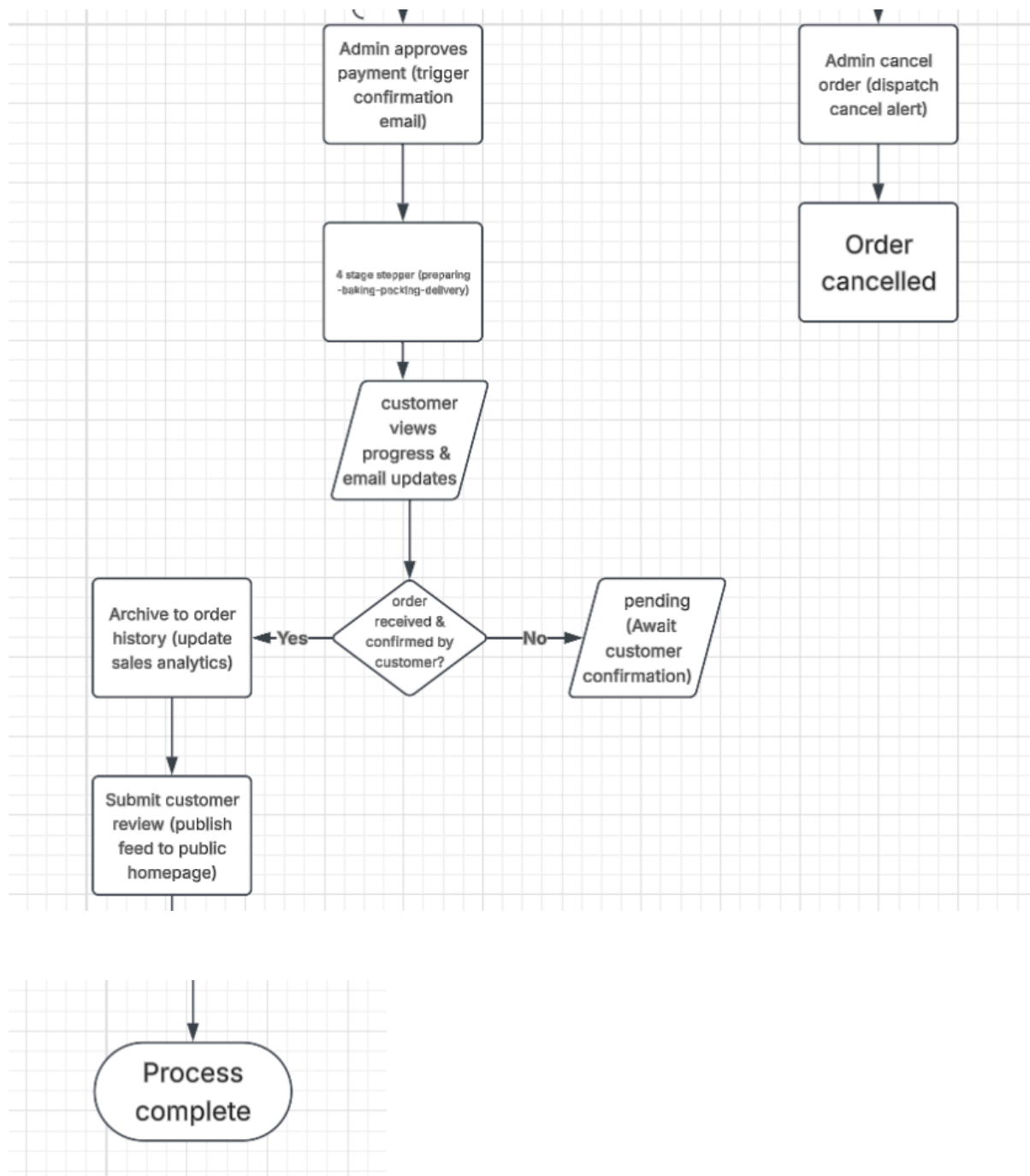
- Automated Cake Ordering allows customers to design custom cakes and submit their specific preferences directly through the website.
- Real-Time Booking Limits automatically check daily capacity during checkout to prevent overbooking on high-demand dates.
- Automated Email Updates send instant notifications to customers whenever their order is approved, updated, or ready for pickup or delivery.
- The 4-Stage Stepper provides store staff with a clear pipeline moving from Preparing, Baking, and Packing to Delivery/Pickup to track and update active orders.
- Live Order Tracking keeps payment statuses and fulfillment progress synchronized between the admin dashboard and the customer tracking page in real time.
- Sales Analytics & Reviews collect completed transaction data to generate business sales reports and store customer feedback to help guide store decisions.

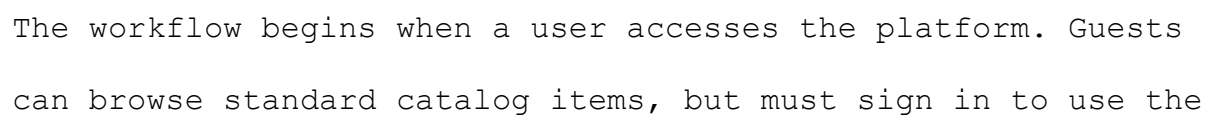
The development of the C&N Bakery System follows modern software engineering principles by incorporating responsive interface design, real-time status synchronization, automated email notification triggers, user authentication, and transaction data processing. The completed system consists of interconnected customer and administrative modules that work together to capture, process, track, and analyze order data.

Figure 4.1 illustrates the overall operational workflow of the developed system.









Custom Cake Builder and cart. In the cart, the user selects their preferred fulfillment date, fulfillment method, and payment option. When the user places the order, the system checks daily schedule capacity in real time; if the selected date is full, an alert prompts the user to select another date. Once submitted, the order is logged as pending. Because the platform operates on the free Firebase database , customers manually send their payment receipt directly to the official C&N Bakery Facebook page. Staff review the payment receipt in Messenger and approve the order in the admin portal. Upon admin confirmation, the system triggers an automated order confirmation email to the customer. Staff then advance the order through 4-Stage Stepper (*Preparing, Baking, Packing, Ready for Pickup/Delivery*). Once delivered, the transaction updates the sales analytics and unlocks customer review submissions.

4.3 System Development Methodology

We chose the Agile Software Development Methodology for this project because building a custom cake ordering platform comes with a lot of moving parts. Features like live price calculations, interactive cake builders, and dynamic date limits usually need tweaking as you test them. Going with Agile gave us the flexibility to adapt quickly whenever we got feedback from the bakery owner, staff, or everyday customers.

Instead of trying to build everything at once, we broke the project down into two-week sprints. Each sprint had a clear goal: early on, we focused on setting up user accounts and customer profiles, then moved on to building the Custom Cake Builder, setting up the payment process, and finally creating the admin controls.

Every sprint started with a quick planning session to decide what to build next. We held short daily check-ins to fix any code issues right away, and at the end of each sprint, we demoed a working version of the app to the bakery team. Working this way helped us catch bugs early, avoid massive re-writes, and build something that actually solves C&N Bakery's day-to-day problems.

The development process was carried out across seven key phases:

Requirements Gathering

The first phase focused on understanding the main challenges in C&N Bakery's daily operations. The team gathered insights through interviews with the bakery owner and staff, observations of their order process, and reviews of paper order slips and sales records. These discussions revealed several key problems with manual workflows, including miscommunications on custom cake designs, inconsistent pricing calculations, difficulties

managing daily booking limits, and a lack of real-time status updates for customers. The insights collected during this stage defined the core requirements for the platform, ensuring the system directly solves the bakery's real-world operational needs.

System Planning

The team set the project scope, timeline, and system specifications. The project was divided into two main parts: a customer portal for building custom cakes, menu catalog, checking date availability, and tracking orders; and an admin dashboard for staff to manage menus, schedules, orders, and sales reports.

A feasibility study confirmed the system was practical to build and run. Technically, the web stack and Firebase Realtime Database could easily handle instant price updates, slot checks, live order tracking, and sales reporting. Operationally, the system was designed to be easy to learn so staff could review payments, approve orders, and manage kitchen queues without slowing down daily tasks. Economically, to stay within Firebase's free 1 GB storage limit, payment screenshots are sent directly to the bakery's Facebook Messenger. This keeps high-volume images off the database, storing only lightweight text data so the system can run reliably with zero hosting costs.

System Design

During this phase, the team created the visual layouts, data structures, and process flows for both the customer and admin portals. An Entity Relationship Diagram (ERD) was mapped out to structure the Firebase database and Schedules—ensuring fast data queries while staying well within the free storage limits. Process flowcharts and UML diagrams standardized key business routines, including custom cake ordering and the 4 stage stepper(*Preparing - Baking - Packing - Ready for Pickup/Delivery*). Additionally, high-fidelity wireframes were designed to build responsive interfaces optimized for both mobile users building custom cakes and staff managing daily orders on desktop screens.

Software Development

The coding phase brought the system design to life using HTML5, CSS3, and Vanilla JavaScript, with Firebase handling the backend service. Visual Studio Code was used for development, while Git and GitHub managed version control across sprints. Firebase Authentication was set up to separate customer accounts from admin privileges and store delivery details. The Custom Cake Builder was coded with a dynamic price estimation engine that automatically recalculates totals as users select tiers, sizes, and design options. Cart logic was built to check real-time

daily capacity limits in Firebase to block overbooking before orders are placed. Finally, state management logic was created for the admin workspace to handle manual payment approvals, advance orders through the 4-stage progress tracker, trigger live updates on the customer dashboard, and render dynamic sales charts.

Testing

Each completed module underwent unit testing before being integrated into the entire system. Integration testing ensured that the interaction between modules—such as cart operations syncing with schedule limits and price calculations reflecting in real time—functioned correctly, while system testing verified the overall performance and security of the application. The Firebase Console Dashboard was continuously monitored during testing to inspect live database traffic, verify read/write security rules, and check real-time system performance logs. User Acceptance Testing (UAT) was later conducted with selected bakery staff and target customers to validate whether the application met user expectations and operational requirements.

Deployment

Once testing was complete, the platform was launched into its live production environment for customers and staff. The backend

was hosted using the free Firebase Spark Plan, with secure role-based rules set up in the database to protect customer data, order histories, and sales reports within the 1 GB storage limit. All image assets such as cake catalog photos, design options, and site graphics were compressed to ensure fast page loading on both mobile and desktop devices. The web application was deployed to secure, HTTPS encrypted hosting to keep all user transactions safe. Finally, admin accounts were created, and initial store configurations including product menus, cake builder pricing, store hours, and daily booking limits were loaded into the database to launch the system for live orders.

Maintenance

The final phase focuses on monitoring platform performance, fixing any reported issues, applying software updates, and improving features based on ongoing user feedback. The team uses the Firebase Console Dashboard to track live database traffic, verify security rules, and monitor database storage usage. Continuous maintenance ensures the platform stays reliable, secure, and helpful for C&N Bakery over time.

Figure 4.2 presents the Agile Software Development Life Cycle adopted in the study.

